

Capital accumulation and growth

The basic Solow model

The previous chapter raised a basic economic question: how can a nation escape from poverty and ultimately become rich? Or more precisely: how can a country initiate a growth process that will lead it to a higher level of GDP and consumption per person?

This chapter presents a fundamental economic model that delivers some first answers. The model will show how the long-run evolutions of income and consumption per worker in a country are affected by structural parameters such as the country's rate of saving and investment and the growth rate of its population. The model is known as the Solow model of economic growth.

In 1956 the economist and Nobel Prize Laureate Robert M. Solow published a seminal article called 'A Contribution to the Theory of Economic Growth'. The article presented a coherent dynamic model with an explicit description of the process of *capital accumulation* by which saving and investment become new capital.¹

In the Solow model, competitive clearing of factor markets implies that output in each period is determined by the available supplies of capital and labour. Furthermore, total saving and investment is assumed to be an exogenous fraction of total income, and the labour force is assumed to grow at a given rate.

The essential additional feature of the Solow model is that it incorporates the dynamic link between the flows of savings and investment and the stock of capital. Solow's model accounts for the fact that between any two successive periods, the stock of capital will increase by an amount equal to gross investment minus depreciation on the initial capital stock.²

The model describes how capital evolves as a result of capital accumulation, how the labour force evolves as a result of population growth, and how total production and income evolves as a consequence of the evolutions of the total inputs of capital and labour. The model therefore involves a certain evolution of income per worker as well. It thereby contributes to answering the fundamental question of what determines 'the wealth of nations'.

This chapter presents the Solow model in its most basic version where the focus is on capital accumulation, not on technological progress, as a source of economic growth.

1. Robert M. Solow, 'A Contribution to the Theory of Economic Growth', *Quarterly Journal of Economics*, **70**, 1956, pp. 65–94.

2. Solow was not the first to suggest a coherent formal framework with an explicit description of the capital accumulation process. Two famous predecessors are Roy. F. Harrod, 'An Essay in Dynamic Theory', *Economic Journal*, **49**, pp. 14–33, and Evsey D. Domar, 'Capital Expansion, Rate of Growth, and Employment', *Econometrica*, **14**, pp. 137–147. However, Solow was the first to suggest such a framework in a model with long-run substitution possibilities between capital and labour. The assumption that these inputs can substitute considerably for each other in the long run is empirically plausible.

Technological progress will be a main theme in coming chapters. Considering the simplicity of the *basic Solow model*, it will take us remarkably far in understanding potential sources of and limitations to economic growth and long-run prosperity: *the basic Solow model* makes a great contribution and is indeed an important workhorse model in economics.

3.1 The basic Solow model

The model presented in this chapter describes a closed economy (the next chapter will present a Solow model for an open economy). Initially we will not explicitly include the public sector in the model, but we will show that it can easily be interpreted as including government expenditure and taxation.

Agents, commodities and markets

The economic agents in the model are households, also referred to as consumers; firms, also called producers; and possibly a government. Time runs in a discrete sequence of periods indexed by t . A period should be thought of as one year. There are three commodities in each period, and for each commodity there is a market. The commodities are output, capital services and labour services.

In the market for output the supply consists of the total output of firms, Y_t . The demand is the sum of total consumption, C_t , and total investment, I_t . Hence output can either be used for consumption, or it can be transformed into capital via investment (at no cost, we assume). The model is thus a one-sector model and does not distinguish between production of consumption goods and production of capital goods. The price in the output market is normalized to one, so other prices are measured in units of output.

We may think of the accumulated stock of physical capital as being directly owned by the households who lease it to the firms. Hence, in the market for capital services the supply comes from the consumers. By a convenient definition of the unit of measurement for capital services (machine years), a capital stock of K_t units of capital can give rise to a supply of K_t units of capital services during period t . The demand for capital services, K_t^d , comes from the firms. The real price, r_t , in this market is the amount of output that a firm must pay to a consumer for leasing one unit of capital during period t . Thus r_t is a real *rental rate*.

This rental rate has a close association with, but is not equal to, the model's real *interest rate*. The reason is that capital depreciates. We assume that the use of one unit of physical capital for one period implies that an amount δ , where $0 < \delta < 1$, must be set aside to compensate for depreciation, that is, for the capital that is worn out by one period's use. The name for δ is the rate of depreciation. The model's real interest rate, ρ_t , is the return to capital *net of depreciation*, that is, $\rho_t = r_t - \delta$. This is the rate of return on capital comparable to an interest rate earned from a financial asset like a bond.

Alternatively, one can think of physical capital as being owned by firms who finance their acquisition of capital by issuing debt to consumers. In the latter case the real price of the use of a unit of capital for one period, also called the 'user cost', is the real interest rate on debt plus the depreciation on one unit of capital. In this interpretation one can think of the firm's 'rental rate' as $r_t = \rho_t + \delta$. For our purposes it does not matter whether the user cost of capital, r_t , is a direct leasing rate or the sum of an interest rate and a depreciation rate.

In the labour market the supply of labour services, L_t , comes from the households, while the demand, L_t^d , comes from the firms. We measure labour flows in man years, so a labour force of L_t can give rise to a labour supply of L_t . The real wage rate in period t is denoted by w_t .

All three markets are assumed to be perfectly competitive, so economic agents take the prices as given, and in each market the appropriate price adjusts so that price-taking supply becomes equal to price-taking demand. This implies that available resources are fully utilized in all periods or, in an alternative interpretation, that they are utilized up to the 'natural rate' defined in Chapter 1.

The production sector

The production side of the economy is modelled as if all production takes place in one representative profit-maximizing firm. The firm uses capital input, K_t^d , and labour input, L_t^d , to produce output (value added), Y_t , according to the production function:

$$Y_t = F(K_t^d, L_t^d). \quad (1)$$

The production function is assumed to display *constant returns to scale*. In mathematical terms this means that the function $F(K^d, L^d)$ is homogeneous of degree one, so $F(\lambda K^d, \lambda L^d) = \lambda F(K^d, L^d)$ for all $\lambda > 0$. In words, if we increase both inputs by, say, 100 per cent, output will also go up by 100 per cent. There is a *replication argument* for this assumption: it should be possible to produce twice as much from a doubling of inputs, since one could simply apply the same production process twice.

The marginal product of capital is the increase in output generated per extra unit of capital. Formally, it is the partial derivative $F_K(K^d, L^d)$ of $F(K^d, L^d)$ with respect to K^d . Similarly, the marginal product of labour is the partial derivative of the production function with respect to L^d , denoted by $F_L(K^d, L^d)$. We assume that F_K and F_L are positive for all input combinations (K^d, L^d) . In accordance with standard microeconomic theory, the marginal products are also assumed to be decreasing in the amount of the factor used, so $F_{KK}(K^d, L^d) < 0$ and $F_{LL}(K^d, L^d) < 0$, where F_{KK} is the second derivative with respect to K^d , and F_{LL} is the second derivative with respect to L^d . Finally, when the input of one factor increases, the marginal product of the other factor is assumed to go up. This means that $F_{KL}(K^d, L^d) = F_{LK}(K^d, L^d) > 0$.

The variables K_t^d and L_t^d are the amounts of capital and labour *demanded* by the firm in period t . The firm faces the real factor prices r_t and w_t , and it will want to choose Y_t , K_t^d and L_t^d to maximize pure profits $Y_t - r_t K_t^d - w_t L_t^d$, subject to the technical feasibility constraint, $Y_t = F(K_t^d, L_t^d)$. From microeconomic theory we know that a competitive profit-maximizing firm will want to employ a factor of production up to the point where its marginal product is just equal to its real price. In our model, the necessary first-order conditions (resulting from differentiation, etc.) take the form:

$$F_K(K_t^d, L_t^d) = r_t, \quad (2)$$

$$F_L(K_t^d, L_t^d) = w_t. \quad (3)$$

Note that (2) and (3) do not determine the levels of K_t^d and L_t^d given r_t and w_t , even though they are two equations in two unknowns. If one combination (K_t^d, L_t^d) fits in the two equations, so does $(\lambda K_t^d, \lambda L_t^d)$ for any $\lambda > 0$. This is a consequence of the assumed homogeneity of the production function.³ Nevertheless, a combination of factor inputs, K_t^d and L_t^d , must fulfil (2) and (3) to be optimal.

Because the factor markets are perfectly competitive, the rental rate, r_t , adjusts to equate capital demand with capital supply, and the wage rate, w_t , adjusts such that labour demand becomes equal to labour supply. We will argue below that as long as the rental rate is just slightly positive, it will pay for consumers to supply all of their capital to the market, so capital supply is equal to the capital stock, K_t . Furthermore, we are going to assume that consumers supply labour inelastically, so labour supply must equal the labour force, L_t . Inserting $K_t^d = K_t$ and $L_t^d = L_t$ into (2) and (3), respectively, gives:

$$F_K(K_t, L_t) = r_t, \quad (4)$$

$$F_L(K_t, L_t) = w_t. \quad (5)$$

3. If a function is homogeneous of degree k , then its first partial derivatives are homogeneous of degree $k - 1$. Since F is homogeneous of degree one, F_K and F_L are homogeneous of degree zero. Hence, $F_K(\lambda K^d, \lambda L^d) = F_K(K^d, L^d)$.

In each period t , the stocks of capital and labour are predetermined and given, as will be explained below. Therefore (4) and (5) determine the period's rental rate, r_t (and hence the interest rate, $r_t - \delta$), and the wage rate, w_t , respectively, as these depend on the given levels of capital and labour. This leads to a theory of the functional distribution of income.

The distribution of income and the Cobb–Douglas production function

Using (1), (4) and (5), we find that the shares of capital and labour in total income in any period t are:

$$\frac{r_t K_t}{Y_t} = \frac{F_K(K_t, L_t) K_t}{F(K_t, L_t)}, \quad (6)$$

$$\frac{w_t L_t}{Y_t} = \frac{F_L(K_t, L_t) L_t}{F(K_t, L_t)}. \quad (7)$$

Note that

$$\frac{F_K K}{Y} \equiv \frac{\partial Y}{\partial K} \frac{K}{Y}$$

is the elasticity of output with respect to the input of capital, and

$$\frac{F_L L}{Y} \equiv \frac{\partial Y}{\partial L} \frac{L}{Y}$$

is the elasticity of output with respect to labour input. Thus equations (6) and (7) say that the share of each factor in total income is equal to the elasticity of output with respect to that factor.

Notice also that the firm's pure profit measured in real terms is $Y_t - (r_t K_t + w_t L_t) = F(K_t, L_t) - [F_K(K_t, L_t) K_t + F_L(K_t, L_t) L_t]$. Since the production function is homogeneous of degree one, we have $F(K_t, L_t) = F_K(K_t, L_t) K_t + F_L(K_t, L_t) L_t$, implying that pure profits are zero. This is an appealing property of a perfectly competitive economy in long-run equilibrium, because if pure profits were positive, we would expect new firms to enter the market until profits were competed away, and if pure profits were negative, we would expect some failing firms to be driven out of the market until the remaining firms were able to break even.

In Chapter 2 we saw that in the long run the factor income shares are remarkably constant. How does this constancy fit with our theory of the income distribution? At first sight it does not seem to fit. According to (7), the labour income share is given as the output elasticity with respect to labour input, and this will generally depend on the input combination, (K_t, L_t) , which is changing over time. More precisely, labour's share depends on the capital–labour ratio, K_t/L_t .⁴ The capital–labour ratio has been systematically increasing over time in developed countries. Apparently this should have implied a changing labour income share, according to our theory.

A similar observation was already puzzling the economist Paul Douglas in 1927. He asked the mathematician Charles Cobb whether a production function exists which has all the properties assumed above, and which will produce constant income shares when factors are paid their marginal products. As we have seen in (6) and (7), this is the same as requiring the output elasticities with respect to capital and labour to be constant (and hence

4. Write labour's share as $F_L(K, L)/F(K, L)$. The numerator only depends on K/L , because F_L is homogeneous of degree zero, so $F_L(K, L) = L^0 F_L(K/L, 1) = F_L(K/L, 1)$, and the denominator only depends on K/L , because F is homogeneous of degree one, so $F(K, L)/L = L F(K/L, 1)/L = F(K/L, 1)$.

independent of K_t/L_t). Charles Cobb found that there exists one such function, since then called the Cobb–Douglas production function:

$$Y_t = F(K_t^d, L_t^d) = B(K_t^d)^\alpha (L_t^d)^{1-\alpha}, \quad B > 0, \quad 0 < \alpha < 1. \quad (8)$$

Here α and B are given parameters, and B is referred to as the total factor productivity, TFP. The TFP is often considered as a measure of technology, but it really captures the influence of all factors other than those explicitly present in the production function (here capital and labour), among these technology.

Note that the inputs of capital and labour are indexed by t , capturing that these amounts of input may vary over time. The total factor productivity, B , has no time subscript, so we have assumed here that there is no change in ‘technology’ over time. This is the sense in which the model in this chapter is the *basic* Solow model. The subject of Chapter 5 will be the Solow model with technological progress where TFP increases over time.

Taking partial derivatives in (8), and inserting the equilibrium conditions, $K_t^d = K_t$ and $L_t^d = L_t$, we find that Eqs (4) and (5) take the particular form:

$$F_K(K_t, L_t) = \alpha B \left(\frac{K_t}{L_t} \right)^{\alpha-1} = r_t, \quad (9)$$

$$F_L(K_t, L_t) = (1 - \alpha) B \left(\frac{K_t}{L_t} \right)^{\alpha} = w_t. \quad (10)$$

Using (8), (9) and (10), you should check carefully that the Cobb–Douglas production function does indeed have *all* the properties attributed to F , and that it does imply constant income shares, a capital share of α , and a labour share of $1 - \alpha$.

The proximate constancy of income shares in the long run suggests that a production function of the Cobb–Douglas form is a reasonable approximation for an aggregate production function for long-run analysis. The fact that the labour income share stays rather close to $\frac{2}{3}$ (see again Chapter 2) suggests that we have a reasonable value for the α appearing in (8), namely a number around $\frac{1}{3}$. Both these features will be used extensively in the growth theory presented in this book. Aggregate production functions will be assumed to have the Cobb–Douglas form, and whenever we need an estimate of α , we will consider $\frac{1}{3}$ as a reasonable value.

The household sector

The number of consumers in period t is L_t , which is predetermined from the past evolution of the population. The behaviour of consumers is described by four features.

First, as mentioned, each consumer is assumed to supply one unit of labour inelastically in each period, so the total supply of labour is L_t . If households were trading off consumption against leisure so as to maximize utility, an inelastic supply of labour would follow if the income and substitution effects from a change in the real wage were exactly offsetting each other. Empirically, long-run labour supply is not too sensitive to changes in real wages, so the assumption of an inelastic labour supply is not a bad first approximation.

Second, the consumers own the capital stock K_t , which is predetermined from capital accumulation in the past. For any positive rental rate, $r_t > 0$, a consumer seeking to maximize his income will want to lease *all* his capital to the firms, because each consumer considers r_t as given. Therefore the supply of capital services in year t is inelastic and equal to the size of the capital stock K_t .

Third, each consumer must decide how much to consume and how much to save. We assume that the household sector behaves as one representative consumer who earns all the

economy's gross income, $Y_t = r_t K_t + w_t L_t$. The representative consumer must decide on consumption, C_t , and hence on gross saving, $S_t \equiv Y_t - C_t$, respecting his intertemporal budget constraint. According to this constraint, the addition, $K_{t+1} - K_t$, to the consumer's real stock of capital from period t to $t + 1$ must equal his gross saving in period t minus the depreciation, δK_t , on the capital leased to the firms during period t . Hence the intertemporal budget constraint is:

$$K_{t+1} - K_t = S_t - \delta K_t. \quad (11)$$

If we were deriving consumption and savings behaviour from explicit optimization, a utility function defined over streams of present and future consumption would be maximized under the intertemporal budget constraint (11). Here, however, we make a simplifying short cut. We assume that the outcome of the underlying optimization is a constant savings rate, i.e. the household sector saves the exogenous fraction, s , of total income in each period:

$$S_t = sY_t, \quad \text{where } 0 < s < 1. \quad (12)$$

We are going to show below that the assumption of a given, constant rate of saving and investment is empirically plausible in the long run. The intertemporal budget constraint, (11), is itself important in our model: it is the aggregate *capital accumulation equation* stating that new capital originates from net savings.

Fourth, the 'biological' behaviour of the households is described by an exogenous growth rate, n , of the population, or rather of the labour force:

$$L_{t+1} = (1 + n)L_t, \quad \text{where } n > -1. \quad (13)$$

The complete model

The basic Solow model consists of the six equations stated in the box below and all repeated from above. Note that as part of the model, initial values, K_0 and L_0 , for the state variables capital and labour are stated in equations (18) and (19).

THE COMPLETE BASIC SOLOW MODEL



$$Y_t = BK_t^\alpha L_t^{1-\alpha}, \quad (14)$$

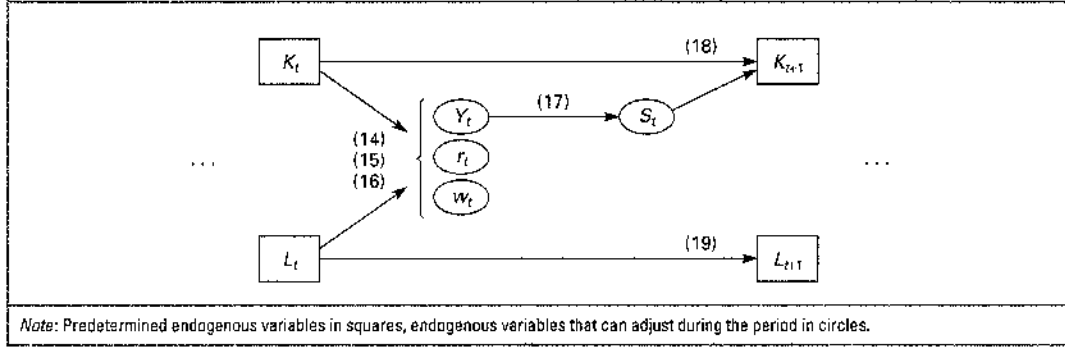
$$r_t = \alpha B \left(\frac{K_t}{L_t} \right)^{\alpha-1}, \quad (15)$$

$$w_t = (1 - \alpha) B \left(\frac{K_t}{L_t} \right)^\alpha, \quad (16)$$

$$S_t = sY_t, \quad (17)$$

$$K_{t+1} - K_t = S_t - \delta K_t, \quad K_0 \text{ given} \quad (18)$$

$$L_{t+1} = (1 + n)L_t, \quad L_0 \text{ given} \quad (19)$$

FIGURE 3.1 The dynamics of the basic Solow model

The parameters of the model are α , B , s , n and δ . Given an initial input combination K_0 , L_0 in some initial year zero, these six equations determine the dynamic evolution of the economy, as illustrated in Fig. 3.1.

In period t the inputs K_t and L_t are predetermined, so the first of the above equations gives the supply-determined output, Y_t , directly while the next two equations give the rental rate, r_t , and the wage rate, w_t , respectively. With Y_t determined, Eq. (17) gives the gross saving, S_t , in period t , and then from S_t and K_t , Eq. (18) determines the next period's capital supply, K_{t+1} . Finally, from the labour supply L_t of period t , Eq. (19) determines the next period's labour supply, L_{t+1} . Hence, K_{t+1} and L_{t+1} have been determined, and one can start all over again with period $t+1$ etc.

We have chosen to present the fundamental *capital accumulation equation*, (18), as the representative consumer's intertemporal budget constraint. We could instead have explained the capital accumulation equation in the following more direct way: by definition, gross investment, I_t , minus depreciation, δK_t , is the addition to capital: $K_{t+1} - K_t = I_t - \delta K_t$. The national accounting identity for the output market, $Y_t = C_t + I_t$, is equivalent to $I_t = S_t$, since $S_t \equiv Y_t - C_t$. Inserting S_t for I_t in the definition of gross investment gives (18).

Government

The model can be interpreted as involving a government sector. Assume that the government collects a tax revenue of T_t in period t , and that government expenditure for public consumption and for public investment are C_t^g and I_t^g , respectively. Let expenditure behaviour be described by the ratio of public consumption to GDP, c_t^g , and by the ratio of public investment to GDP, i_t^g , respectively:

$$C_t^g = c_t^g Y_t, \quad \text{where } 0 < c_t^g < 1,$$

$$I_t^g = i_t^g Y_t, \quad \text{where } 0 < i_t^g < 1.$$

We allow c_t^g and i_t^g to be time dependent and assume $c_t^g + i_t^g < 1$. Assume further that the government balances its budget:

$$T_t = C_t^g + I_t^g.$$

Let c_t^p denote the ratio of private consumption, C_t^p , to private disposable income, $Y_t - T_t$. Then:

$$C_t^p = c_t^p (Y_t - T_t), \quad \text{where } 0 < c_t^p < 1.$$

We also allow c_t^p to be time-dependent.⁵ Combining the above equations one finds that:

$$C_t^p = c_t^p(1 - c_t^g - i_t^g)Y_t.$$

Thus private consumption as a fraction of GDP is $c_t^p(1 - c_t^g - i_t^g)$, which is not necessarily a constant. Total (national) consumption, $C_t = C_t^g + C_t^p$, is:

$$C_t = [c_t^g + c_t^p(1 - c_t^g - i_t^g)]Y_t.$$

The assumption we make is that the share of *total* consumption in GDP, $c_t^g + c_t^p(1 - c_t^g - i_t^g)$, is a given constant. Then national savings must be a constant fraction of GDP as well, $S_t = sY_t$, where:

$$s = 1 - c_t^g - c_t^p(1 - c_t^g - i_t^g) = (1 - c_t^p)(1 - c_t^g) + c_t^p i_t^g. \quad (20)$$

Again, national gross investment must equal national gross saving, $I_t = S_t$. Hence, all we need to do to interpret a government into our model (14)–(19) is to think of S_t as the sum of private and public saving and to think of the constant saving and investment rate, s , appearing in (17) as being given by underlying parameters as in (20). Note that s , the national investment rate, is decreasing in c_t^p and c_t^g , and increasing in i_t^g (explain, for instance, why i_t^g affects s through the term $c_t^p i_t^g$).

An essential assumption underlying the Solow model is that the sum of the GDP shares of private and public consumption, which is the share of total consumption in GDP, is a given constant. Figure 3.2 indicates that the ratio of total consumption to GDP has indeed stayed relatively constant over long periods in developed economies, with a tendency for C_t^p/Y_t and C_t^g/Y_t to move in opposite directions, keeping the sum, C_t/Y_t , relatively stable.

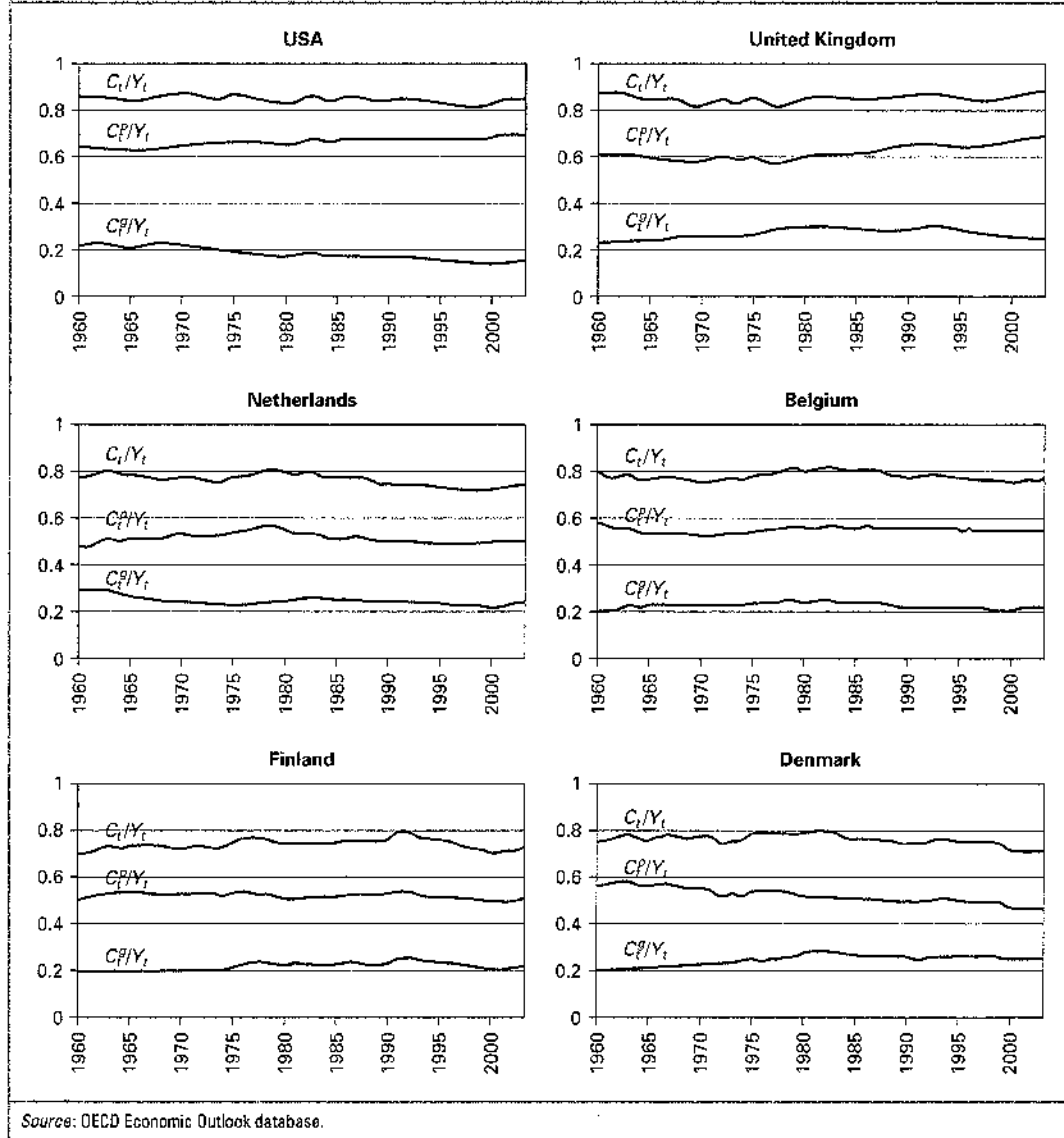
Table A at the end of Book One collects some data relevant for growth analysis, among them average (gross) investment rates (s_k in the table) over the period 1960 to 2003 for a large number of countries. You will see that values around, or somewhat above, 20 per cent are typical for developed countries, while less developed countries can have investment rates far below 20 per cent. For a typical Western developed country, a reasonable value for the model parameter s could thus be, say, 22 per cent.

We should emphasize that sometimes breaks in national saving and investment rates occur. For instance, Fig. 3.2 gives an indication that the Netherlands and Denmark experienced downward structural breaks in the consumption share of GDP in the early 1980s. We are not arguing that savings rates stay constant for ever, only that they tend to stay sufficiently constant over sufficiently long periods to justify a modelling assumption of an exogenously given savings and investment rate. Indeed, one important economic event that we want to be able to analyse in our growth model is a permanent structural change in the savings rate from one constant value to another.

Money?

There is no direct trace of ‘money’ in the Solow model. This is because most economists think that *in the long run money does not matter* for real economic variables. One should treat this statement with caution. Certainly the quality of the financial system can have a long-run impact, for instance by affecting our parameter s , as we will explain later in this chapter. Likewise, many economists think that the quality of monetary stabilization policy, that is, the systematic way that monetary authorities respond to economic events, can have economic consequences also in the long run, cf. the discussion in Chapter 1 of the

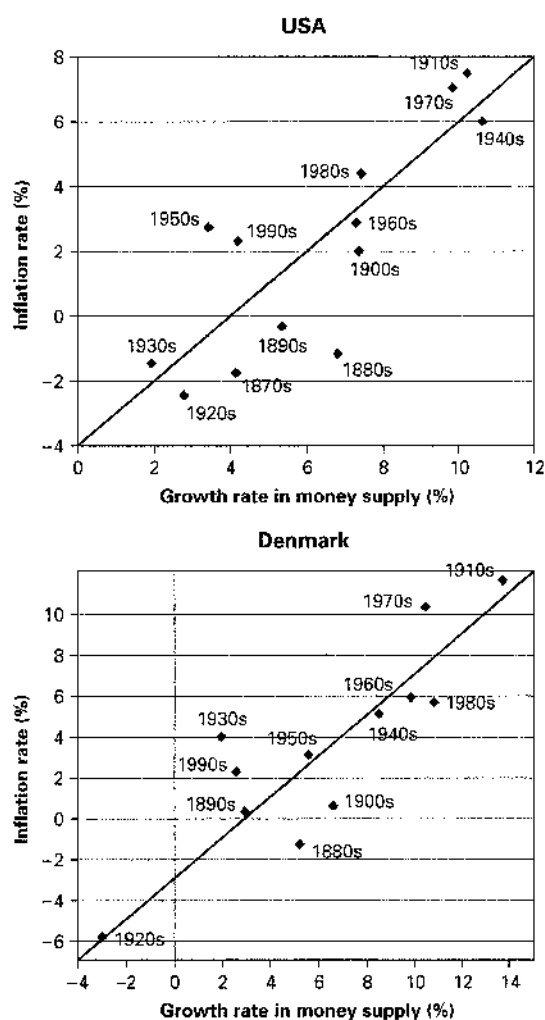
5. In Chapter 15 we shall see that c_t^p is actually quite stable in the long run.

FIGURE 3.2 The consumption share of GDP in several Western countries

relationship between stability in the short run and growth in the long run. However, given the quality of the financial system, most economists think that the exact level or growth rate of the nominal money stock, or of the short-term nominal interest rate set by the central bank, do not affect real variables such as GDP in the long run.

Figure 3.3 gives a sharp illustration. Both in the USA and in Denmark a change in money growth of a given size goes hand in hand with a change in inflation of the same size in the longer run (note that the figure focuses on the long run by considering averages over periods of 10 years).

The figure does not *prove* that money growth has no real impact in the long run. There is, however, a rather sophisticated econometric literature on this issue. In his advanced textbook on monetary economics, the economist Carl E. Walsh gives an account of this

FIGURE 3.3 Money growth and inflation, Denmark and the USA, 1870–2000

Note: Money growth and inflation rates are average annual rates over each decade. Money supply is measured by M2.

Sources: For the USA: M. Friedman and A. Schwartz, *A Monetary History of the United States 1867–1960*, The University of Chicago Press, 1982 and IMF International Financial Statistics. For Denmark: Niels Kærgård, *Økonomisk Vækst: En Økonometrisk Analyse af Danmark 1870–1981*, Jurist-og Økonomforbundets Forlag, 1991 and OECD Economic Outlook (through EcoWin).

literature and summarizes as follows as regards the long run:⁶ ‘Money growth and inflation display a correlation of 1; the correlation between money growth or inflation and real output growth is probably close to zero...’. On the other hand, the conclusion ‘...on the short-run effects of money is that exogenous monetary policy shocks produce hump-shaped movements in real economic activity. The peak effects occur after a lag of several quarters (as much as two or three years in some of the estimates) and then die out’.

This should explain why you will hear no more of money in Book One of this text, while monetary policy will be at the heart of the analysis throughout Book Two.

6. Carl E. Walsh, *Monetary Theory and Policy*, Cambridge and London, MIT Press, 2000.

Time

In one respect our Solow model does not quite look as suggested by Solow himself. We have formulated time as running discretely like the integers 0, 1, 2, 3, ... Perhaps it is more in accordance with how most people think of time to assume that time runs smoothly, or continuously, in an uninterrupted stream like the real numbers.

Often the Solow model is formulated in continuous time. In that case flow variables such as production, consumption, saving and investment are smooth functions of time to be interpreted as intensities. For instance, if the investment intensity at time t is $I(t)$, then total investment over the year from time $t = 0$ to $t = 1$ is $\int_0^1 I(t) dt$, so if the investment intensity stays constant at I over the full year, then the total investment during the year is I . Stock variables are also smooth functions of time with the obvious interpretation that $K(t)$, for instance, is the total capital stock at time t .

In continuous time the change in capital must be expressed as a time derivative,

$$\frac{dK(t)}{dt} \equiv \dot{K}(t) = \lim_{\Delta t \rightarrow 0} \frac{K(t + \Delta t) - K(t)}{\Delta t},$$

which measures the change in capital per unit of time (since there is a Δt in the denominator) at a specific *point* in time (at a nanosecond, so to speak). So, $\dot{K}(t)$ is the intensity of change in capital at time t . The counterpart of the discrete time expression, $K_{t+1} - K_t = I_t - \delta K_t$, is $\dot{K}(t) = I(t) - \delta K(t)$, stating that the intensity of change in capital at time t is the gross investment intensity minus the depreciation intensity at time t . When formulating the Solow model in continuous time, one often drops the explicit reference to time and one simply writes $\dot{K} = I - \delta K$, since the dot indicates change over time anyway.

With these conventions, the Solow model in continuous time consists of the following six equations, the counterparts of (14)–(19) above. The first one, for instance, says that the production intensity at any time is given as a Cobb–Douglas function of the intensities by which capital and labour are used at that time. The last one says that the growth rate of population is n at all times:

THE SOLOW MODEL IN CONTINUOUS TIME



$$Y = BK^\alpha L^{1-\alpha}, \quad (21)$$

$$r = \alpha B \left(\frac{K}{L} \right)^{\alpha-1}, \quad (22)$$

$$w = (1-\alpha)B \left(\frac{K}{L} \right)^\alpha, \quad (23)$$

$$S = sY, \quad (24)$$

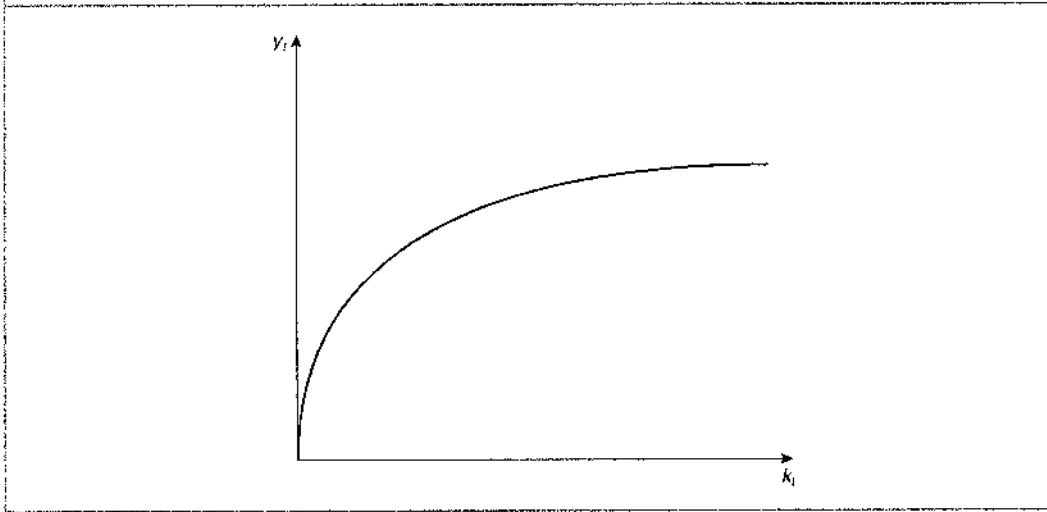
$$\dot{K} = sY - \delta K, \quad K_0 \text{ given} \quad (25)$$

$$\frac{\dot{L}}{L} = n, \quad L_0 \text{ given} \quad (26)$$

The exercises at the end of the chapter will ask you to analyse the Solow model in continuous time. This will help you to master both ways of modelling time, which will be important and useful in your further studies.

It is a matter of choice whether to formulate the Solow model in discrete time or in continuous time. The main argument in favour of continuous time is that it is sometimes mathematically more convenient. The case for discrete time modelling is that it is directly applicable to empirical data which are typically reported for discrete periods like a year or a quarter, and it is easier to do simulation exercises with a discrete time model on a computer.

FIGURE 3.4 The per capita production function



3.2 Analysing the basic Solow model

Output and capital per worker

For the prosperity of a nation it is GDP per worker or per capita, not GDP itself, that is important. In the Solow model we are therefore interested in output per worker, $y_t \equiv Y_t/L_t$. Define similarly capital per worker in period t , also called the capital–labour ratio, or the capital intensity: $k_t \equiv K_t/L_t$. From the first equation of the Solow model, (14) above, it follows from dividing on both sides by L_t that:

$$y_t = Bk_t^\alpha. \quad (27)$$

This is like a production function where output is GDP per worker and input is capital per worker. This production function exhibits diminishing returns with the marginal product, $\alpha Bk_t^{\alpha-1}$, decreasing from infinity to zero as k_t increases from zero to infinity. The *per capita production function* is illustrated in Fig. 3.4.

One feature revealed directly by (27) is that in the basic Solow model (where B is assumed to be constant), an increase in production per worker can only come from an increase in capital per worker. Let the approximate growth rates of y_t and k_t from period $t-1$ to period t be denoted by g_t^y and g_t^k , respectively. Writing an equation like (27) for both period t and period $t-1$, taking logs on both sides, and subtracting gives:

$$g_t^y \equiv \ln y_t - \ln y_{t-1} = \alpha(\ln k_t - \ln k_{t-1}) \equiv \alpha g_t^k. \quad (28)$$

According to (28), the (approximate) growth rate in output per worker is proportional to the growth rate in capital per worker, and the proportionality factor is exactly the capital share, α . In the basic Solow model economic growth is thus completely linked to capital accumulation.

The law of motion

We can analyse the Solow model directly in terms of the variables we are interested in, the per capita (or worker) magnitudes, k_t and y_t . First, insert the savings behaviour (17) into the capital accumulation equation (18) to get:

$$K_{t+1} = sY_t + (1-\delta)K_t.$$

Then divide this equation on both sides by L_{t+1} , on the right-hand side using that $L_{t+1} = (1+n)L_t$. Using also the definitions of k_t and y_t , one gets:

$$k_{t+1} = \frac{1}{1+n} [sy_t + (1-\delta)k_t].$$

Finally, from the per capita production function, (27), we can substitute Bk_t^α for y_t to arrive at:

$$k_{t+1} = \frac{1}{1+n} [sBk_t^\alpha + (1-\delta)k_t]. \quad (29)$$

This is the basic law of motion, or the *transition equation*, for the capital intensity k_t following from the basic Solow model. Note that (29) has the form of a first-order, one-dimensional, non-linear difference equation. For a given initial value, k_0 , of the capital labour ratio in year zero, (29) determines the capital-labour ratio, k_1 , of year one, which can then be inserted on the right-hand side to determine k_2 , etc. In this way, given k_0 , the transition equation (29) determines the full dynamic sequence (k_t) of capital intensities.⁷

The four equations (14) and (17)–(19), which gave the dynamic evolution of Y_t , S_t , K_t and L_t , have thus been boiled down to the single equation (29) giving the dynamic evolution of k_t . With (k_t) determined, the factor prices will be given from (15) and (16):

$$r_t = \alpha Bk_t^{\alpha-1}, \quad (30)$$

$$w_t = (1-\alpha)Bk_t^\alpha. \quad (31)$$

The sequence for output per worker, (y_t) , follows from (k_t) by $y_t = Bk_t^\alpha$. Total consumption in period t is $C_t = Y_t - S_t = (1-s)Y_t$, so consumption per worker is $c_t = (1-s)y_t$. Therefore (c_t) follows from (k_t) via $c_t = (1-s)Bk_t^\alpha$. In this way the dynamic evolution of all the variables of interest can be derived, given an initial capital intensity.

There is another illustrative way to state the economy's law of motion, where one expresses the *change* in the capital intensity as a function of the current capital intensity. Subtracting k_t from both sides in (29) gives the so-called *Solow equation*:

$$k_{t+1} - k_t = \frac{1}{1+n} [sBk_t^\alpha - (n+\delta)k_t]. \quad (32)$$

This has a nice intuitive interpretation. On the left-hand side is the increase in capital per worker. On the right-hand side are the elements that can create such an increase: saving per worker in period t is $sy_t = sBk_t^\alpha$, which adds to capital per worker, depreciation per worker in period t is δk_t , which subtracts from capital per worker, and finally there is a subtraction, nk_t , which is the thinning-out of capital per worker caused by the fact that there are more workers in period $t+1$ to share a given capital stock (if $n > 0$).

You may wonder why n enters (32) exactly as it does. Here is an explanation. Assume that the capital stock is unchanged from t to $t+1$: $K_{t+1} = K_t$. This means that capital per worker will change from t to $t+1$ by an amount caused by population growth, and by that alone. Using the definition of k_t , we have in that case: $k_{t+1}/k_t = (K_{t+1}/K_t)(L_t/L_{t+1}) = 1/(1+n)$, so $k_{t+1} = k_t/(1+n)$, and then $k_{t+1} - k_t = -[n/(1+n)]k_t$. This shows how capital per worker evolves when there is no compensation for population growth. You should recognize the way n enters (32).

7. The notation (k_t) will be used for the full dynamic sequence for k_t , that is, (k_t) lists k_t for $t = 0, 1, 2, \dots$

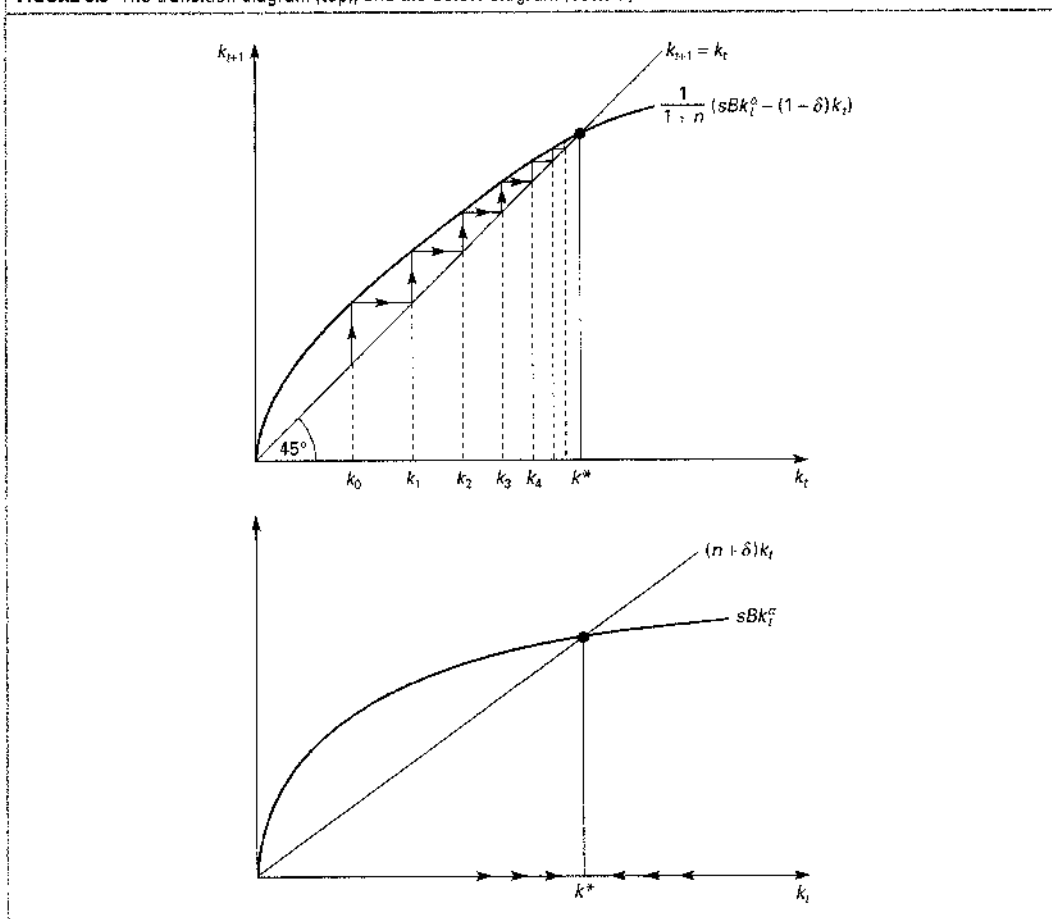
FIGURE 3.5 The transition diagram (top), and the Solow diagram (bottom)**Convergence to steady state**

Figure 3.5 illustrates the dynamics of the Solow model. The upper part of the figure, the *transition diagram*, shows the transition equation as given by (29). This curve starts at $(0, 0)$ and is everywhere increasing. The 45° line, $k_{t+1} = k_t$, has also been drawn.

Differentiating (29) gives:

$$\frac{dk_{t+1}}{dk_t} = \frac{s\alpha Bk_t^{\alpha-1} + (1-\delta)}{1+n}.$$

This shows that the slope of the transition curve decreases monotonically from infinity to $(1-\delta)/(1+n)$, as k_t increases from zero to infinity. The latter slope is positive and less than one if $n+\delta > 0$, or $n > -\delta$, which is plausible empirically. Usually annual depreciation rates on capital are estimated to be between 5 and 15 per cent (there is some uncertainty here because there are many kinds of capital with different depreciation rates), while most often the lower end of the interval around 5 per cent is considered most plausible for *aggregate* capital. Annual population growth rates never slump to -0.05 (see, for instance, Table A at the end of Book One). We can therefore safely assume $n > -\delta$, implying that the slope of the transition curve falls monotonically from infinity down to a value less than one. Hence the

transition curve must have a unique intersection with the 45° line (which has slope one), to the right of $k_t = 0$.

The lower part of Fig. 3.5, the so-called *Solow diagram*, illustrates the Solow equation (32). It contains the curve sBk_t^α , and the ray $(n + \delta)k_t$. The vertical distance between the first and the second, divided by $(1 + n)$, is $k_{t+1} - k_t$ according to (32). The two curves must intersect each other once to the right of $k_t = 0$, since the ray has constant slope $n + \delta > 0$, and the slope of the curve falls monotonically from infinity to zero. Where this intersection occurs, we have $k_{t+1} - k_t = 0$, or $k_{t+1} = k_t$. Hence the intersection must be at the same k_t at which the transition curve intersects the 45° line in the upper part of the figure.

The iterations of (29) described above appear in the transition diagram as follows. Assume an initial capital intensity $k_0 > 0$, as drawn. Then k_1 will be the vertical distance from the horizontal axis up to the transition curve, and by going from the associated point on the transition curve horizontally to the 45° line, and then vertically down, k_1 will be taken to the horizontal axis as shown. Now, k_2 will be the vertical distance up to the transition curve. The dynamic evolution of the capital intensity is given by the staircase formed of broken lines. It follows that over time, k_t will converge to the specific value k^* given by the intersection between the transition curve and the 45° line, and it will do so monotonically, getting closer and closer all the time and never transcending to the other side of the intersection point. Furthermore, the convergence is *global* in the sense that it holds for any strictly positive initial k_0 .⁸ Of course, as k_t converges to k^* , income per worker, $y_t = Bk_t^\alpha$, will converge to its steady state value, $y^* = Bk^*$.

In the Solow diagram, when the curve is above the ray, one must have $k_{t+1} - k_t > 0$, or $k_{t+1} > k_t$, so k_t must be increasing over the periods, and vice versa. To the left of the intersection point, k_t is therefore increasing over time towards k^* , while to the right it is decreasing towards k^* , as indicated by the arrows along the k_t axis. Summing up:

CONVERGENCE TO STEADY STATE



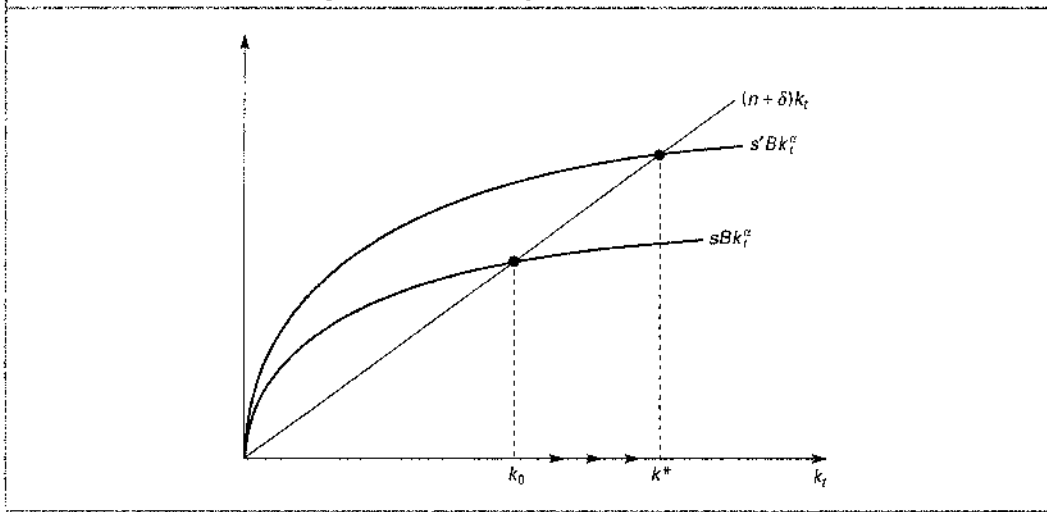
Under the plausible stability condition, $n + \delta > 0$, the basic Solow model implies that the capital intensity (capital per worker) converges monotonically to a specific value k^* . This k^* is called the steady state value for the capital intensity. At the same time income per worker converges to a corresponding steady state value, y^* .

We have described the model's convergence to steady state in mathematical terms. The steady state is important and Section 3.3 below will focus on it. Before turning to that, we want to illustrate the properties of our model a bit further using the Solow diagram. This will, among other things, explain the convergence process in economic terms.

Comparative analysis in the Solow diagram

Figure 3.6 is a Solow diagram just like the lower part of Fig. 3.5. It contains the ray $(n + \delta)k_t$ and two different curves, sBk_t^α and $s'Bk_t^\alpha$, where $s' > s$. We imagine that the economy has first been characterized by the parameter values α , B , s , n and δ for a long time, but then from some year zero, the gross savings rate shifts to s' and stays at its new higher level thereafter, while no other parameters change. Remember that since $s = S_t/Y_t = I_t/Y_t$, we may as well say that it is the gross investment rate that has increased.

8. Mathematically, to establish global, monotonic convergence we have been using the following properties of the transition curve: it has a unique intersection with the 45° line for $k_t > 0$, everywhere to the left of this intersection (except in $k_t = 0$) it lies above the 45° line and everywhere to the right it lies below, and its slope is everywhere positive.

FIGURE 3.6 An increase in the savings rate in the Solow diagram

Since the old parameter values have prevailed for a long time, the economy should initially be at (or close to) the steady state corresponding to the old parameter values, k_0 in the figure. In this steady state current savings are just large enough to compensate capital per worker for depreciation and population growth, so k_t stays unchanged from period to period.

In year zero, when the increase in the savings rate occurs, the initial capital intensity will be unchanged, k_0 , because it is predetermined and given by capital accumulation and population dynamics in the past, where the savings rate was s . Therefore output per worker stays unchanged at $y_0 = Bk_0^\alpha$ as well. In the short run, on impact, output and income per worker are thus unaffected. However, out of the unchanged income per worker, y_0 , a larger part will now be saved, $s'y_0$ in period zero against sy_0 previously. Current savings will therefore more than compensate for depreciation and population growth. In the figure, $s'Bk_0^\alpha > (n + \delta)k_0$, so capital per worker will increase from period zero to period one. With $k_1 > k_0$, one will also have $y_1 > y_0$, and out of the increased income per worker will come even more savings per worker, $s'Bk_1^\alpha > s'Bk_0^\alpha$. Hence capital per worker will increase again, $k_2 > k_1$, and so on. In the long run successive increases in the capital intensity and in output per worker will make k_t converge to the new and higher steady state value k^* , and y_t will converge to the new and higher steady state value $y^* = B(k^*)^\alpha$. These increases reflect the economy's increased capacity for saving.

One can use the Solow diagram for another type of comparative analysis which does not involve a parameter shift. As above, assume that the economy is initially in steady state at parameters α , B , s , n and δ , so the capital intensity is k_0 , etc. Suppose that the economy's capital per worker then drops to $\frac{1}{3}$ of its old value at the beginning of period one, say, because most of the capital is destroyed in a war or an earthquake during period zero, while the parameters stay unchanged. (This may illustrate something like what happened to Japan in the Second World War.) We can analyse this situation in the Solow diagram by keeping the curves in it unchanged, but reducing the capital intensity to $(\frac{1}{3})k_0$ from period one. You should do this in a Solow diagram and convince yourself that the economy will immediately start a recovery back towards its initial steady state. Perhaps this recovery can partly describe the strong growth performance of Japan after the Second World War. In an exercise you will be asked to analyse further the remarkable recovery of the Japanese economy in the postwar period and investigate whether this seems to accord with the Solow model.

3.3 Steady state

Our model predicts that in the long run the capital intensity and GDP per worker converge to particular steady state levels, k^* and y^* , respectively. It is of interest to express k^* , y^* and other key variables of the steady state as functions of the model's parameters α , B , s , n and δ . This will show us which fundamental characteristics of an economy can create a high level of income and consumption per worker in the long run according to the basic Solow model. Furthermore, it will enable us to confront the model's steady state predictions with the data.

The key endogenous variables in steady state

The steady state capital–labour ratio is given as the unique constant solution, $k_{t+1} = k_t = k$, to (29) or (32) above. From the latter, such a k must fulfil: $sBk^\alpha - (n + \delta)k = 0$, or $k^{1-\alpha} = sB/(n + \delta)$. Hence:

$$k^* = B^{\frac{1}{1-\alpha}} \left(\frac{s}{n + \delta} \right)^{\frac{1}{1-\alpha}}. \quad (33)$$

Given the assumption $n + \delta > 0$, this is a meaningful expression.⁹ Output per worker in steady state is found by inserting the particular value k^* for k_t in the per capita production function, $y_t = Bk_t^\alpha$:

$$y^* = B^{\frac{1}{1-\alpha}} \left(\frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}}. \quad (34)$$

Consumption per worker is $c_t = (1 - s)y_t$ in any period t , so:

$$c^* = B^{\frac{1}{1-\alpha}} (1 - s) \left(\frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}}. \quad (35)$$

Likewise, inserting k^* into (30) and (31) gives the real factor prices in steady state:

$$r^* = \alpha \left(\frac{s}{n + \delta} \right)^{-1}, \quad (36)$$

$$w^* = (1 - \alpha) B^{\frac{1}{1-\alpha}} \left(\frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}}. \quad (37)$$

These simple and innocent-looking formulae contain some very sharp and important predictions about an economy's long-run performance. For instance, (34) gives a handy formula for how an economy's GDP per worker should depend on a few basic structural parameters, such as the rate of investment and the population growth rate, in the long run. Taking logs on both sides of (34) gives:

$$\ln y^* = \frac{1}{1-\alpha} \ln B + \frac{\alpha}{1-\alpha} \ln s - \frac{\alpha}{1-\alpha} \ln(n + \delta), \quad (38)$$

9. The fact that we have found exactly one positive solution confirms our earlier statement that there is a unique positive intersection between the transition curve and the 45° line in the transition diagram, or the curve and the ray in the Solow diagram.

from which, for instance, the elasticity of y^* with respect to s is $\alpha/(1 - \alpha)$, and the elasticity with respect to $n + \delta$ is $-\alpha/(1 - \alpha)$ (show this by differentiation on (38)). You will easily verify that the elasticity of w^* with respect to s is also $\alpha/(1 - \alpha)$ etc. Since α is around $\frac{1}{3}$, the elasticity, $\alpha/(1 - \alpha)$, should be around $\frac{1}{2}$.

What can make a country rich in the long run? According to the basic Solow model, the answer is:

LONG-RUN INCOME PER CAPITA



A relatively high technological level, B , a high rate of gross saving and investment, s , a low population growth rate, n , and a low depreciation rate, δ , all tend to imply a relatively high GDP per worker in the long run according to the basic Solow model. The model predicts that the elasticity of long-run income per capita and of long-run real wages with respect to s should be around one half, and the elasticity with respect to $n + \delta$ should be minus one half, so, for instance, a 20 per cent increase in the investment rate, e.g. from 20 to 24 per cent, should imply an increase in people's average incomes of around 10 per cent in the long run.

The 'natural' interest rate: productivity and thrift

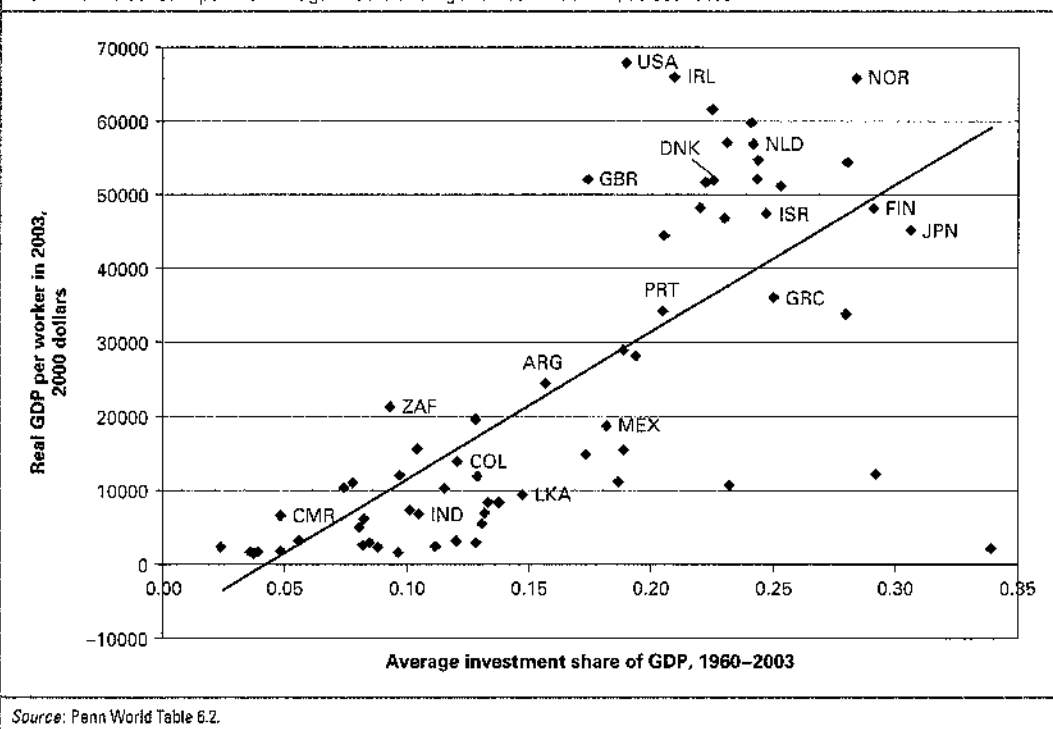
Just as (34) gives a prediction for the long-run GDP per worker, (36) gives a prediction for the real interest rate. Recall that the r^* of (36) is the steady state value for the rental rate for capital, while the steady state real interest rate is $\rho^* = r^* - \delta$, so:

$$\rho^* = \alpha \frac{n + \delta}{s} - \delta. \quad (39)$$

The ρ^* of (39) is sometimes referred to as the 'natural' rate of interest, following the Swedish economist Knut Wicksell who was the first to use this term.¹⁰ We see that the natural rate of interest in the basic Solow model is determined by the forces of *productivity* and *thrift*. The parameter α measures the response of output to an increase in capital input. The higher the value of α , the higher is the productivity of capital, and the greater will be the demand for capital, and hence for replacement investment to compensate for depreciation and population growth in steady state. A higher value of $n + \delta$ increases the demand for replacement investment as well, and hence increases the equilibrium real interest rate, the price of capital. This is the influence of productivity on ρ^* . The influence of thrift is reflected in the appearance of s : the higher the national propensity to save, the greater the supply of capital, and the lower the equilibrium real interest rate, *ceteris paribus*.

As argued above and from our Table A, reasonable values by Western standards for the parameters entering into (39) could, on an annual basis, be: $\alpha = \frac{1}{3}$, $s = 0.22$, $n = 0.005$, and $\delta = 0.05$. You will easily compute that these parameters result in a value for the natural interest rate per annum of $\rho^* = 3.3$ per cent. This is very close to the long-run average values of annual real interest rates implied by the observed historical difference between nominal interest rates on long-term bonds and the observed rates of inflation. Although we are going to revise our formula for the natural real interest rate somewhat in more sophisticated models later on, it is promising that our extremely simple model predicts a long-run real interest rate that is so close to empirical estimates.

10. Knut Wicksell wrote at the turn of the twentieth century. Today he is considered as one of the great figures in the history of economic thought. His theory of interest rates is summarized in: Knut Wicksell, 'The Influence of the Rate of Interest on Prices', *Economic Journal*, 17, 1907, pp. 213–220.

FIGURE 3.7 Real GDP per worker against the average investment share, 65 countries

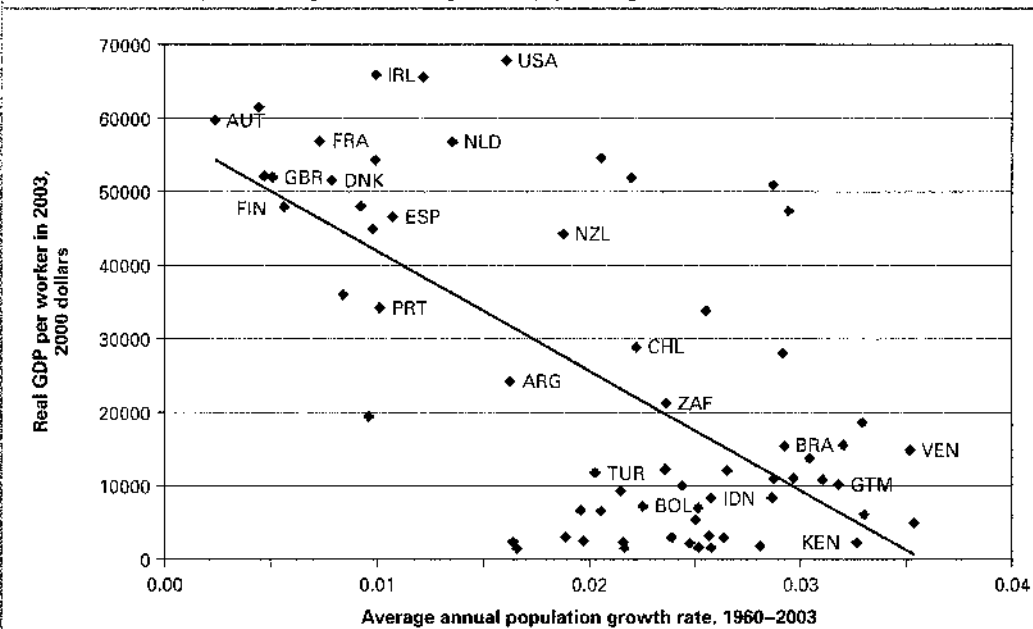
Source: Penn World Table 6.2.

Testing the model's steady state prediction of GDP per worker

We can test our prediction of long-run GDP per worker empirically, for instance by considering cross-country data such as those in Table A. Here we run into the difficulty that it is not so easy to get reliable data for differences between countries in technological variables such as B and δ . However, we do have good data for GDP per worker and for the investment and population (or labour force) growth rates for many of the countries in the world, as evidenced by Table A. How well does the prediction of (34) fit with the data on these variables?

Figure 3.7 plots the GDP per worker in 2003 against the average gross investment rate between 1960 and 2003 across countries for the representative sample of 65 countries that we also studied in Figs 2.5 and 2.6. The selection criteria for the sample were described when we presented those figures. Basically the sample consists of countries that are not too small or too dependent on oil, and for which we have data of good quality from the Penn World Table 6.2. Figure 3.8 plots GDP per worker in 2003 against average labour force growth rates between 1960 and 2003 for the same countries. The increasing relationship in Fig. 3.7 and the decreasing one in Fig. 3.8 are in nice accordance with the basic Solow model. A high savings rate and a low labour force growth rate do seem to go hand in hand with a high level of GDP per worker.

Later we will discuss the goodness of these fits further. In particular, we will investigate whether the model prediction of a long-run elasticity of output with respect to $\ln s - \ln(n + \delta)$ of around $\frac{1}{2}$ stands up against the empirical evidence (and we ask you to do part of the job in the important Exercise 9 at the end of this chapter). For now we conclude:

FIGURE 3.8 Real GDP per worker against the average annual population growth rate, 65 countries

Source: Penn World Table 6.2.

THE BASIC SOLOW MODEL EMPIRICALLY

Cross-country empirical evidence is consistent with the long-run predictions of the basic Solow model that a higher saving/investment rate and a lower growth rate of the labour force tend to increase income per capita.

This is promising for a research agenda building on the Solow model and it is the main reason we claimed in the introduction to this chapter that the model takes us remarkably far in understanding the sources of long-run prosperity.

3.4 Structural policy and the golden rule

Crowding out

Consider the model interpretation with a government, where the national savings and investment rate, s , is given by (20). Assume that government increases its propensity to consume, c_t^g , such that s decreases permanently to a new and lower constant level as given by (20). What will be the effects of such a policy according to our model starting from a steady state?

This is easy to answer since we have already studied the effects of an increase in s , and the effects are now just the opposite. Hence, in the short run (in the first period of the new and lower s), where K_t is predetermined by capital accumulation in the past according to the old investment rate, both K_t and L_t , and hence Y_t , will be unaffected by the policy change: Y_t will continue to grow at the rate n , ensuring that GDP per worker stays unchanged in the first period. This should remind you of lessons learned earlier, for example, in connection with the so-called classical macroeconomic model: in a period where output is determined

from the supply side by the available amounts of resources, an increase in government demand has no influence on GDP, but private investment will be fully crowded out by the increase in government consumption.

In the long run, the policy change considered *will* have an effect on GDP and on GDP per capita. Out of the (unchanged) output and income in the first period of the new policy, a smaller fraction than before will be saved and invested, and therefore there will be less capital, and hence output, in the succeeding period, leading to a further fall in saving and capital accumulation, etc. Over time Y_t will thus develop more slowly than without the policy change, thus bringing output per worker to a new and lower steady state value, reflecting the economy's decreased potential for saving and investing. In the long run there is *more* than full crowding out, a result arising from the explicit consideration of the capital accumulation process. The decrease in capital supply caused by the government's increased propensity to consume also implies that the long-run equilibrium interest rate increases, as we noted in connection with (39) above.

It is an important and general insight that a government cannot create a positive influence on GDP or GDP per capita *in the long run* by boosting government consumption. However, it may be able to do so *in the short run* if the economy is not currently at the capacity constraint. In that case a rise in public consumption will lead to less than full crowding out of private spending, as we shall see in Book Two on short-run economic fluctuations and stabilization policies. Here we are concerned with the long run and with structural policies.

Should one conclude from the above analysis that in the long run government expenditure can have only a negative influence on economic activity and consumer welfare, because of its negative influence on capital accumulation? Certainly not, and for a number of reasons.

Government expenditure motivated by long-run concerns

First, instead of increasing its spending on consumption, the government could choose to spend an increase in tax revenues on investment in, say, infrastructure. In our model this possibility is represented by an increase in i_t^g rather than in c_t^g , and an increase in i_t^g has a positive influence on s , as shown by (20). Thus, a tax-financed increase in government's propensity to invest raises the national investment rate to a new and higher level, which has a positive long-run impact on GDP per worker.

Second, consumers may be willing to accept a fall in aggregate national income arising from an increase in government consumption if this increase enables the public sector to provide essential public goods which cannot be provided in adequate quantities by private, profit-motivated producers via the marketplace.

Third, if public consumption spent on, say, education and health is a near-perfect substitute for private spending on these items, a tax-financed increase in public spending on schools and hospitals will most likely reduce private consumption on education and health by a roughly similar amount, leaving total saving and investment, and thereby total output, unaffected.

Fourth, there may be distributional reasons for wanting part of, for example, health services to be publicly provided or financed. For educational services, both distributional concerns about equality of opportunity and the presence of positive externalities (e.g. the benefits to other citizens that a person is literate) may motivate public provision or financing.

Finally, much of what is normally categorized as public consumption may be seen as delivery of productive inputs to the private sector production process. For example, health and education services may improve the productivity of the labour force, and public institutions guaranteeing law and order (examples of essential public goods) may also be productivity-enhancing and encourage a higher private savings rate by safeguarding property rights. Thus part of public consumption may help to increase the productivity variable B as well as the private savings rate $1 - c_t^p$ in the Solow model.

Incentive policies

We have considered policies that influence the national savings rate, s , through the government's direct control over public consumption and investment. In addition, we can imagine that the government can influence the private savings rate (or the population growth rate or the depreciation rate, for that matter) through incentive policies, for instance policies that restructure the tax system. However, our model does not give any explicit description of the underlying incentive mechanisms.

Whether government is assumed to influence the model's basic parameters one way or the other, a main conclusion is that a government wishing to promote its citizens' long-run incomes should follow structural policies that can somehow improve technology, increase the national propensity to save and invest, and reduce population growth.

Certainly many countries use various programmes to enhance technology, and such activities may be meaningful according to the Solow model. However, if some technology is only adopted by private firms because it is subsidized, there should, from an economist's point of view, be a positive external effect associated with the technology, that is, a positive spillover effect on agents other than the firm investing in the technology. More advanced technology is not preferable at any cost. We will return to the discussion of policies to improve technology later in connection with models with endogenous technological innovation.

Policies to promote *private* savings can be of many types. Quite often these policies take the form of tax incentives for savings and investment. As mentioned, we cannot conduct an explicit analysis of such incentive policies, since we have not derived private savings behaviour from optimization. We can, however, point to some main issues. For example, is capital income overtaxed compared to labour income? Are the different types of capital income taxed in a uniform manner so as to prevent distortions in the pattern of savings and investment?

At the more basic level, a system of well-defined, secure property rights and a sound financial system are important prerequisites for large-scale private savings. There must be safe banks and assets in which savings can be placed. For some poor countries this is a serious issue, and the most essential policy to promote savings in such countries is the establishment of safe property rights and a good financial system. To achieve such a goal may be very difficult if, for instance, there is civil war in the country.

Because of its emphasis on capital accumulation, the basic Solow model strongly suggests that a good system for handling savings and channelling them into productive investment is important for prosperity. This is an example of a general and fundamental economic insight: institutions are very important for economic performance, and in some countries an improvement of institutions is the number one need for initiating positive economic development.

In Chapter 6, when we have extended the Solow model with technological development and human capital, we give a more explicit exposition of 'the importance of institutions'.

The golden rule of saving

One may get the impression from our discussion that it is good to have as high a savings and investment rate as possible. Equation (34) certainly says that a higher s gives higher long-run income per worker, y^* . However, it does not require much reflection to see that a maximal savings rate is not preferable. If s is close to one, GDP per worker will be maximal, but according to (35), consumption per worker will then be close to zero. If the ultimate purpose of production is to enjoy the highest possible level of *consumption* per worker, what should the savings rate be? Using (35) to maximize c^* with respect to s gives the savings rate: $s^{**} = \alpha$ (Exercise 7 will ask you to show and explain this). This consumption-maximizing savings rate, s^{**} , is called the *golden rule* savings rate. The capital income share, α , is around $\frac{1}{3}$. If policy makers are interested in consumption per person, they should not try to drive savings

rates above $\frac{1}{3}$. Perhaps they should not even go close to that level, since the marginal increase in c^* obtained from an increase in s is small if s is already close to α (explain this), and since higher saving implies postponed consumption which represents a welfare cost if people have 'time preference', preferring to consume now rather than later.

As we notice from Table A at the end of Book One, many rich countries in the world have savings and investment rates somewhat above 20 per cent. Some of the East Asian 'growth miracle' countries had investment rates up to around $\frac{1}{3}$ at the end of the twentieth century. However, poor countries typically have much lower savings rates, down to around 10 per cent, and sometimes even considerably less than that. For the latter countries policies that could work to increase savings rates would seem to be important for raising the long-run level of economic welfare.

3.5 Economic growth in the basic Solow model

The macroeconomic dynamics outside steady state in the Solow model are of interest since this is where there can be economic growth.

Transitory growth

The long-run prediction of the Solow model is the steady state, and in steady state there is no growth in GDP per worker. There can be growth in the GDP itself, but only at the speed of population growth.

Why is it that growth has to cease in the long run according to the basic Solow model? Here, diminishing returns to capital plays a crucial role. The production function for output per worker, $y_t = Bk_t^\alpha$, has a 'marginal product', $\alpha Bk_t^{\alpha-1}$, which decreases with k_t because $\alpha < 1$. Assume that the economy is initially below steady state, $k_t < k^*$. Production and income per worker will be Bk_t^α in period t , and the resulting gross savings per worker, sBk_t^α , will exceed the amount $(n + \delta)k_t$ needed to maintain capital per worker in the face of population growth and depreciation (look again at the Solow diagram in the lower part of Fig. 3.5). Hence capital per worker will increase from one period to the next. As k_t increases, each unit of additional capital per worker will generate ever smaller increases in income and gross savings per worker (due to diminishing returns to capital). At the same time the additional savings needed to compensate for population growth and depreciation increase, since k_t is increasing. Therefore, ultimately (in steady state) gross savings per worker, sBk_t^α , will just cover what is needed to keep capital per worker unchanged, $(n + \delta)k_t$. In the Solow diagram diminishing returns is reflected by the function sBk_t^α being curved with a decreasing slope. (Try to envisage the growth process if sBk_t^α were instead a ray lying above $(n + \delta)k_t$, corresponding to $\alpha = 1$, which would mean constant returns to capital alone.)

As the steady state is approached (from below), both k_t and y_t will be growing. Hence, in the basic Solow model there is *transitory growth* in GDP per worker on the way to steady state, but no growth in steady state. During the transition phase, growth in k_t goes hand in hand with growth in y_t . As noted earlier, with the assumed aggregate production function, growth in GDP per worker has to be rooted in growth in capital per worker.

An example

It is relevant to investigate how fast or slow transitory growth is according to the basic Solow model. If growth fades within a few years, the model could not deliver much of an explanation of growth over decades.

Consider an economy that is initially described by the following parameter values: $B = 1$, $\alpha = \frac{1}{3}$, $\delta = 0.05$, $n = 0.03$ and $s = 0.08$. The value of B is just a normalization, while α and δ have been set at reasonable values, for δ assuming a period length of one year. The values for n and s , respectively, correspond to a high annual population growth rate and a low propensity to save, which could be descriptive of a typical poor country (see again

TABLE 3.1 Simulation output

Year	k_t	y_t	c_t	g_t^y , exact (%)*	g_t^y , approx (%)**
0	1.000	1.000	0.920		
1	1.000	1.000	0.760	0.00	0.00
2	1.155	1.049	0.797	4.93	4.81
3	1.310	1.094	0.832	4.28	4.19
4	1.463	1.135	0.863	3.76	3.69
5	1.614	1.173	0.892	3.33	3.27
6	1.762	1.208	0.918	2.97	2.92
7	1.907	1.240	0.942	2.66	2.63
8	2.048	1.270	0.965	2.40	2.38
9	2.184	1.298	0.986	2.18	2.16
10	2.317	1.323	1.006	1.98	1.97
⋮	⋮	⋮	⋮	⋮	⋮
∞ (new st.st.)	5.196	1.732	1.316	0	0

Note: * computed as $(y_t - y_{t-1})/y_{t-1} \times 100$; ** computed as $(\ln y_t - \ln y_{t-1}) \times 100$.

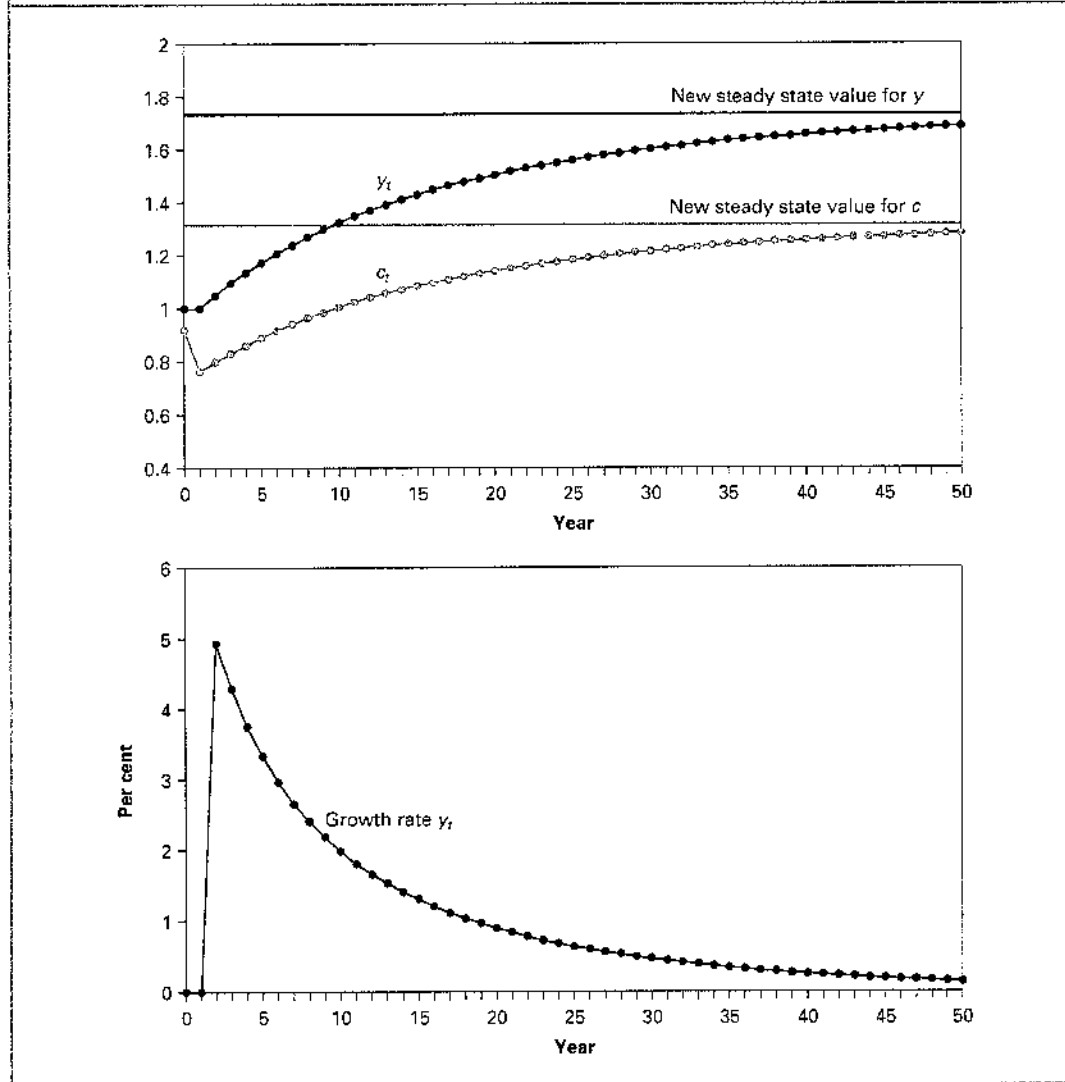
Table A). We assume that the economy is initially in steady state at these parameter values. Using (33) and (34) one finds steady state capital per worker and GDP per worker, respectively: $k^* = y^* = 1$.

Now suppose that from an initial year one, the savings rate increases permanently up to a level typically seen in rich countries, say, $s' = 0.24$, while the other parameters stay unchanged. The new steady state is easily computed: $k^{*'} = 5.20$, and $y^{*'} = 1.73$. In the long run, the very substantial increase in the savings rate of 200 per cent implies a 420 per cent increase in capital per worker and a 73 per cent increase in GDP per worker. (Try to explain why the increase in y^* is not 100 per cent, as an elasticity of y^* with respect to s equal to $\frac{1}{2}$ would suggest.) But how fast is the transition to the new steady state?

We can simulate the transition equation (29) with the savings rate equal to s' and the other parameters as given above over a number of periods $t = 2, 3, \dots$, starting with $k_0 = k_1 = 1$. (In year one capital per worker will be at the old steady state level, since the increased savings rate during period one only has an effect on capital per worker from period two.) In this way a sequence, (k_t) , can be constructed. From (k_t) one can derive a corresponding sequence (y_t) by using $y_t = k_t^\alpha$ in all years, and a sequence (c_t) by using $c_0 = (1 - s)y_0$ and $c_t = (1 - s')y_t$ for $t \geq 1$. For each year one can then compute the growth rate in GDP per worker, $y_t/y_{t-1} - 1$, or approximate by $g_t^y = \ln y_t - \ln y_{t-1}$. These operations are most easily done on a computer using a spreadsheet.

The simulation gives rise to the time series shown in Table 3.1, and illustrated graphically in Fig. 3.9, which depicts the evolution in y_t , c_t and g_t^y over some 50 years after the change in the savings rate.

There are several interesting features to note from Fig. 3.9. It illustrates the growth perspectives, according to the Solow model, for a poor country bringing its savings rate up to the standards of rich countries. For instance, there is a substantial initial sacrifice of consumption for some years, six to be precise (explain this). Furthermore, there is considerable transitory growth for a long time. It takes around 12 years to get half the way from the old to the new steady state level of GDP per capita. After 20 years the annual growth rate of GDP per worker is still close to 1 per cent, and this is still purely transitory growth arising from the permanent increase in the savings rate 20 years earlier.

FIGURE 3.9 The evolution of y_t , c_t and g_t^y after the increase in s 

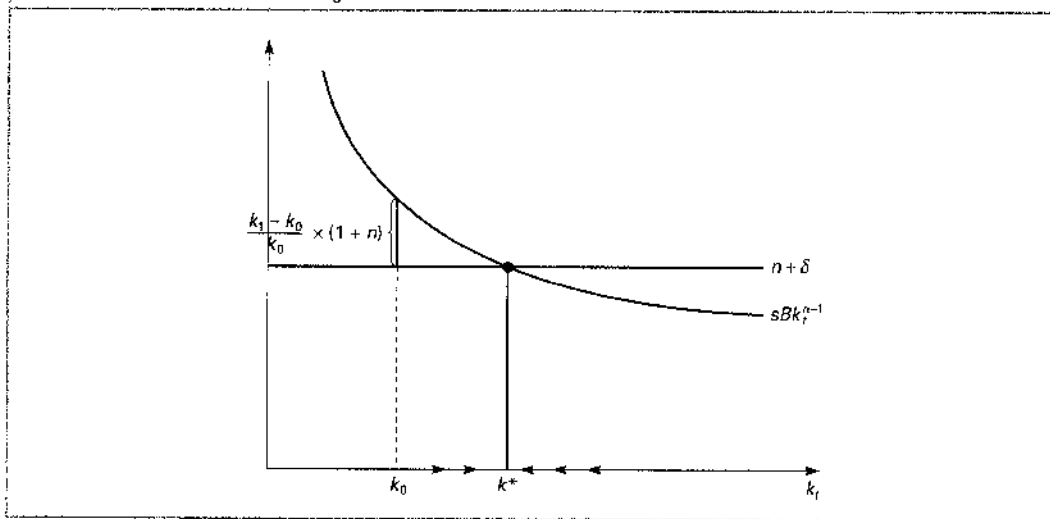
Transitory growth in the Solow model takes place over decades for realistic parameter values. This relative slowness in the adjustment outside steady state is why it is justified to say that: *the basic Solow model is a growth model*.

A 'cousin' to the Solow diagram is useful for illustrating the growth process outside steady state. Dividing by k_t on both sides of (32) gives the *modified Solow equation*:

$$\frac{k_{t+1} - k_t}{k_t} = \frac{1}{1+n} [sBk_t^{\alpha-1} - (n+\delta)]. \quad (40)$$

In the modified Solow diagram, Fig. 3.10, the growth rate of the capital intensity is thus proportional to the vertical distance between the decreasing curve, $sBk_t^{\alpha-1}$, and the horizontal line at $n+\delta$. The steady state is, of course, where the curve and the line intersect.

FIGURE 3.10 The modified Solow diagram



It follows that the growth rate in k_t will be higher (in absolute value) the further away from the steady state k_t is. The same will be true for the growth rate in y_t , since $g_t^y = \alpha g_t^k$. In the numerical example above this feature appeared. After the initial increase in the savings rate, the growth rate of y_t jumped up and then decreased monotonically back towards zero.

The model feature that growth is higher the further below steady state the economy is accords with the idea of conditional convergence discussed in Chapter 2. In a modified Solow diagram such as Fig. 3.10, the positions of the curve and the line depend on the structural characteristics of the economy. If one country has a higher savings rate than another, the countries being otherwise structurally alike, the decreasing curve of the first country will be situated above that of the second country. If the two countries start out with the same GDP per worker, the first country will grow faster and converge to a higher final level of GDP per worker. However, *conditional* on countries being structurally alike, the Solow model predicts that countries will converge to the same level of GDP per worker through a growth process where the countries that are initially most behind will grow the fastest. In other words, the Solow model supports conditional convergence.

THE CONVERGENCE PROCESS



For realistic parameter values the transitory growth of the basic Solow model is relatively slow and takes place 'over decades'. The further below steady state GDP per capita the economy starts out, the higher is its growth rate. This is an empirically appealing feature of the model because it accords with the idea of conditional convergence between the countries of the world.

The long-run increase in income per capita resulting from an increase in the investment rate can be viewed as modest. It also comes slowly and with a considerable consumption sacrifice. However, a central theme in chapters to follow will be that the effect of the investment rate on income per capita seems to be larger than predicted by the basic Solow model.

3.6 Summary



1. The basic Solow model studied in this chapter is characterized by the following features:
 - Aggregate output, GDP, is produced from aggregate capital and aggregate labour according to a Cobb–Douglas production function with constant returns to capital and labour and constant total factor productivity.
 - Competitive market clearing ensures that the amounts of capital and labour used in production in each period are equal to the predetermined available amounts and that factors are rewarded at their marginal products. An implication is that the income shares of capital and labour are equal to the elasticities of output with respect to capital and labour, respectively.
 - Capital evolves over time such that from one year to the next, the change in the capital stock equals gross saving (or investment) minus depreciation on the initial capital stock.
 - In each period gross saving is an exogenous fraction of total income or GDP. Depreciation on capital is given by an exogenous depreciation rate. The labour force changes over time at an exogenous growth rate.
2. The key assumptions of a Cobb–Douglas aggregate production function and a given rate of saving and investment are motivated empirically by the relative constancy of income shares in the long run and the relative long-run constancy of the GDP share of total consumption (private plus public), respectively.
3. The model implies that capital per worker and output per worker converge to particular constant values in the long run. The convergence point defines the economy's steady state. In steady state, consumption per worker, the real interest rate, and the wage rate are constant as well. The steady state has some empirical plausibility with respect to the predicted value for the long-run real interest rate and with respect to the long-run dependence of GDP per worker on the investment rate and on the labour force growth rate.
4. The expressions for the steady state values of the key variables contain some sharp predictions of how income per worker and consumption per worker depend on underlying parameters in the long run. These predictions are of importance for the design of structural policies to raise the standard of living. According to the model, policies to make a nation richer should mainly be policies that can increase the investment share of GDP and bring population growth under control, or policies to improve technology. To create a high level of consumption per person, the savings and investment rate should not exceed the golden rule value, which is equal to the elasticity of output with respect to capital.
5. In the model's steady state there is no positive growth in GDP per worker, consumption per worker, or the real wage rate. This is at odds with the stylized facts of growth (in developed economies). Remedying this shortcoming is one of the main purposes of the growth models to be presented later. These models will contain some form of technological progress, such that total factor productivity increases over time. The basic Solow model does give rise to (positive or negative) growth in output per worker during the transition to steady state. This transitory growth is relatively long-lasting for plausible parameter values. Furthermore, the process of transitory growth is such that the further below steady state an economy currently is, the faster it will grow. This is in accordance with the observed conditional convergence between the countries of the world.



6. Economics has much more to say about the sources of prosperity and growth than what can be inferred from the basic Solow model. For instance, human capital (the education and training embodied in people), and natural resources such as land, oil and metals are factors of production which could be as important as physical capital and 'raw' labour. Furthermore, the evolution of technology has to be an important determinant for long-run prosperity and growth as well. These factors will be dealt with in later chapters, but the basic Solow model is important as it stands. It is therefore worthwhile first modifying the model to take account of a fact we have neglected in this chapter, namely the openness of real-world economies. This openness implies that capital can flow between countries at least to some extent. A Solow model taking capital mobility appropriately into account and describing the accumulation of capital and national wealth in an open economy will be presented in the next chapter.

3.7 Exercises

Exercise 1. The elasticity of substitution between capital and labour and the income distribution

The Cobb–Douglas production function, $Y = BK^\alpha L^{1-\alpha}$ (omitting time subscripts and superscripts d for 'demand') implies a certain degree of substitution between the inputs. As a measure of this degree economists use the so-called elasticity of substitution. This is defined as the elasticity in the composition, K/L , of input demands with respect to the ratio, r/w , between the rental rates, when the firm maximizes profits (minimizes costs) given r and w .

1. Find the marginal rate of substitution between capital and labour (the marginal product of capital divided by the marginal product of labour) for the Cobb–Douglas production function. Show that profit maximization given r and w requires that this marginal rate of substitution is equal to r/w , and show that the elasticity of substitution between capital and labour is -1 , independent of r and w .

The so-called CES (constant elasticity of substitution) production function is:

$$Y = \left(\alpha K^{\frac{\sigma-1}{\sigma}} + (1-\alpha)L^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad 0 < \alpha < 1, \sigma > 0, \neq 1,$$

where α and σ are parameters.

2. Find the marginal rate of substitution between capital and labour for the CES production function, and show that the elasticity of substitution is $-\sigma$, independent of r and w .

The CES function is more general than the Cobb–Douglas function, since the particular value $\sigma = 1$ (or the limit for $\sigma \rightarrow 1$) corresponds to the Cobb–Douglas function. Nevertheless, we have chosen to work with the more special Cobb–Douglas function, partly for simplicity, but also for the reason appearing from the answers to the next two questions.

3. Show that with the CES production function, the income share of labour under competitive market clearing is:

$$\frac{wL}{Y} = \frac{1-\alpha}{\alpha \left(\frac{K}{L} \right)^{\frac{\sigma-1}{\sigma}} + (1-\alpha)}.$$

4. In Chapter 2 it was shown that developed economies have experienced relatively constant positive growth rates in GDP per worker (annual rates around 2 per cent) and relatively constant capital–output ratios in the long run. Show that developed countries must then

also have experienced increasing capital–labour ratios in the long run. If K/L increases and eventually goes to infinity, how should labour's share evolve over time in the case of $\sigma < 1$, $\sigma = 1$ and $\sigma > 1$, respectively, according to the above formula? This gives an argument for considering $\sigma = 1$ most plausible in models for the long run. Explain why.

Exercise 2. The basic Solow model in continuous time

This exercise asks you to analyse the Solow model in continuous time as given by the six equations (21)–(26). The restrictions on the parameters α , B , s , n and δ are the same as in the model in discrete time, except we do not have to assume $n > -1$. It is assumed that $n + \delta > 0$. We use (for any point t in time) the definitions $k = K/L$ and $y = Y/L$ etc.

1. Show that from $k = K/L$ it follows $\dot{k}/k = \dot{K}/K - \dot{L}/L$. (Hint: You can do the ‘log-dif-trick’, that is, first take logs on both sides of $k = K/L$ and then differentiate with respect to time.) Then show that the Solow model in continuous time implies the Solow equation:

$$\dot{k} = sBk^\alpha - (n + \delta)k. \quad (41)$$

Compare with (32) for the model in discrete time and give an intuitive explanation like the one given for (32) in this chapter.

2. Illustrate the above Solow equation in a Solow diagram with k along the horizontal axis and the curve sBk^α as well as the ray $(n + \delta)k$ along the vertical axis. Demonstrate (from the Solow diagram) that from any initial value, $k_0 > 0$, the capital intensity, k , will converge to a specific steady state level, k^* , in the long run (as $t \rightarrow \infty$).
3. Compute the steady state values, k^* , y^* and c^* , for the capital–labour ratio, income per worker, and consumption per worker, respectively (use $\dot{k} = 0$). Explain and compare to the parallel expressions found in this chapter for the model in discrete time. Also compute the steady state values, r^* and w^* , for the real rental rates for capital and labour and compare again to the model in discrete time. How are r^* and w^* affected by an increase in s ? Explain. What are the growth rates of y and Y in steady state?
4. Show that (it follows from the model that) the growth rate in k at any time is:

$$\frac{\dot{k}}{k} = sBk^{\alpha-1} - (n + \delta). \quad (42)$$

Illustrate this in a modified Solow diagram. Show that the growth rate, $\dot{y}/y$, at any time t is α times the growth rate, $\dot{k}/k$, of k . Assume that initially $k_0 < k^*$. Describe and explain qualitatively, in words, the transition to steady state. For instance, how do r and w develop along the transition, and when is the growth rate in k , and hence in y , largest?

Exercise 3. An analytical solution to the Solow model in continuous time

It was stated in this chapter that continuous time is sometimes mathematically more convenient than discrete time. For instance, although the differential equation (41) is not linear, we can actually solve it, that is, we can find the (closed form) function (for k as a function of t) that follows from (41), or from (42), and an initial value k_0 . In the following assume for simplicity that $B = 1$. The trick is to define an auxiliary variable z , $z \equiv k^{1-\alpha}$. First show that z is actually the capital–output ratio, k/y . Then show by doing appropriate substitutions in (42), that:

$$\dot{z} = (1 - \alpha)s - \lambda z, \quad (43)$$

where $\lambda \equiv (1 - \alpha)(n + \delta)$. Note that (43) is linear. Show that the steady state value of z is $z^* = (1 - \alpha)s/\lambda = s/(n + \delta)$. Solve (43) by finding the characteristic roots, etc., and show that (given an initial capital–output ratio z_0) the solution is:

$$\begin{aligned}
 z &= \frac{(1-\alpha)s}{\lambda} (1 - e^{-\lambda t}) + z_0 e^{-\lambda t} \\
 &= \frac{s}{n+\delta} (1 - e^{-\lambda t}) + z_0 e^{-\lambda t}.
 \end{aligned}$$

In other words, at any time t , the capital–output ratio is the weighted average of its steady state value and its initial value, the weight going more and more to the first of these. Show that the solutions for k and y are:

$$\begin{aligned}
 k &= \left(\frac{s}{n+\delta} (1 - e^{-\lambda t}) + k_0^{1-\alpha} e^{-\lambda t} \right)^{\frac{1}{1-\alpha}}, \\
 y &= \left(\frac{s}{n+\delta} (1 - e^{-\lambda t}) + y_0^{\frac{1-\alpha}{\alpha}} e^{-\lambda t} \right)^{\frac{\alpha}{1-\alpha}}.
 \end{aligned}$$

From the expressions verify that the larger λ is, the faster will convergence to steady state be, so that $\lambda \equiv (1-\alpha)(n+\delta)$ can be viewed a measure of the rate of convergence to steady state.

Exercise 4. The effects of a decrease in the population growth rate

1. Explain the economic effects according to the basic Solow model of a decrease (at some time) in the population growth rate from one constant level to a new and lower constant level: how does this affect the transition diagram and the Solow diagram? How does it affect the steady state values of capital, income and consumption per worker? Explain qualitatively the transition from the old steady state to the new one: what initiates the growth process, and what keeps it alive for some time?

Now consider the same parametrization of the basic Solow model as in the example of Section 3.5, that is, $B = 1$, $\alpha = \frac{1}{3}$, $\delta = 0.05$, $n = 0.03$, and $s = 0.08$. Assume that in period zero the economy is in steady state at these parameter values, and then (working first time between period zero and period one) the population growth rate drops to 0.01, other parameters remaining unchanged. (This is illustrative for a developing country getting population growth down to Western rates.)

2. By how many per cent does the reduction by $\frac{2}{3}$ in the population growth rate increase the steady state values of capital, income and consumption per worker, respectively?
3. Carry out a simulation of the model over the periods 0 to 100, and show in a diagram the evolution of y_t , c_t , and the growth rate $(y_t - y_{t-1})/y_{t-1}$ (or the approximate one $\ln y_t - \ln y_{t-1}$), over the 100 periods. Comment with respect to what a developing country can obtain by birth control according to the Solow model. Does there seem to be any systematic difference with respect to the speed of convergence to the new steady state (or the speed by which the benefits arrive) between a decrease in the population growth rate as considered here, and an increase in the savings rate as considered in the example of section 5? For instance, for the two experiments (approximately from the figures) compare the number of periods it takes until half of the distance between the old and the new steady state values for y has been eliminated. Try to explain the difference (Exercise 3 may be useful).

Exercise 5. The effects of an increase in technology and the propagation mechanism

1. Do all you were asked to do in the previous exercise (there for a decrease in n) only this time for a *permanent* increase in B . First analyse the qualitative effects using the Solow diagram and explaining the transition mechanisms. Then consider a numerical example (e.g. $B = 1$, $\alpha = 1/3$, $\delta = 0.08$, $n = 0$, $s = 0.2$) and analyse by appropriate simulations the effects of a permanent increase in B from 1 to 1.2 taking place in period one, assuming the economy was in steady state up to period zero.

2. Now do the same for a *temporary* increase in B , where B goes up from 1 to 1.2 in period one and then falls permanently back to 1. Explain why GDP (per capita) increases compared to the old steady state also in the periods after period one. (Hint: the effect is referred to as propagation by capital accumulation.)

Exercise 6. A one-shot increase in L_t

Analyse in the Solow diagram the qualitative effects of a one-shot increase in L_t , starting in steady state at the prevailing parameter values. The event we are interested in here is that at the beginning of some period, period one say, L_t increases once and for all by a certain amount, for instance due to emigration. However, both before and after this event the population growth rate stays unchanged at n , say. How does this affect the Solow diagram and the capital accumulation process?

Exercise 7. Golden rule

In this chapter it was stated that the golden rule savings rate that maximizes steady state consumption per person as given by (35) is $s^{**} = \alpha$. Show this. To illustrate, draw a Solow diagram with the curve, sBk_t^α , and the ray, $(n + \delta)k_t$, as usual, and draw the curve Bk_t^α as well. For any given k_t , consumption per worker in period t is the vertical distance between Bk_t^α and sBk_t^α . Explain this. The particular steady state capital intensity is where sBk_t^α and $(n + \delta)k_t$ intersect. Hence, *in steady state* consumption per worker is the vertical distance between Bk_t^α and $(n + \delta)k_t$. Compute the particular value k^{**} of k_t that maximizes this distance. Illustrate the point in your Solow diagram: the slope of Bk_t^α should equal the slope of $(n + \delta)k_t$. Explain this. What is the requirement on s to ensure that the steady state capital intensity, k^* , becomes equal to the golden rule capital intensity, k^{**} ? Illustrate again in your diagram.

Exercise 8. Balanced growth in the steady state

Show that *all* the requirements for balanced growth, as these are stated in Chapter 2, are fulfilled in the steady state of the basic Solow model. Why are we, nevertheless, not completely satisfied with the basic Solow model's steady state as a description of long-run growth?

Exercise 9. Further empirically testing the steady state prediction of the basic Solow model

This exercise anticipates some issues taken up in succeeding chapters, and it is a very useful empirical exercise. The data you will need can be taken from Table A. In Figs 3.7 and 3.8 we tested the steady state prediction of the Solow model by plotting, across 65 countries, GDP per worker in 2003 against investment rates (averaged over 1960 to 2003) and population growth rates (also averaged over 1960 to 2003). We concluded that the directions of the empirical relationships were nicely in accordance with the basic Solow model. However, the steady state prediction of the Solow model is more precise than ' y^* is increasing in s and decreasing in n ': as shown by (38), there should be a linear relationship between $\ln y^*$ and $\ln s - \ln(n + \delta)$, and the slope in this relationship should be $\alpha/(1 - \alpha)$.

Test this by creating a figure that plots $\ln y^i$ against $\ln s^i - \ln(n^i + \delta)$ across the same 65 countries i (indicated in Table A), using for y^i , s^i and n^i the same data that were used in Figs 3.7 and 3.8 (appearing in Table A), and setting $\delta = 0.075$. Does the relationship (by reasonable standards) seem to be a linear one? Estimate (by OLS) and draw the line of best fit through the points. What is the estimated slope of this line? How does this slope accord with the theory?

Exercise 10. The recovery of Japan after the Second World War

Table 3.2 shows GDP per worker for the USA and Japan for the period 1959 to 1996, measured in a common, purchasing-power-adjusted currency (1996 dollars). The table only goes to 1996 to avoid considering (in the present long-run context) the more recent years of economic crisis in Japan.

TABLE 3.2 GDP per worker: USA and Japan (1995 dollars)

Year	USA	Japan
1959	29,876	6,880
1960	30,304	7,889
1961	30,260	8,577
1962	31,520	9,164
1963	32,370	9,800
1964	33,710	10,782
1965	35,363	11,167
1966	36,900	12,184
1967	37,037	13,402
1968	38,302	15,047
1969	39,020	16,495
1970	38,432	18,098
1971	39,023	18,736
1972	40,477	20,119
1973	42,184	21,314
1974	41,367	20,735
1975	39,735	21,017
1976	41,469	21,681
1977	42,976	22,462
1978	44,583	23,510
1979	45,167	24,645
1980	44,217	25,252
1981	44,756	25,743
1982	42,741	26,254
1983	44,048	26,532
1984	47,159	27,353
1985	48,164	28,368
1986	49,271	28,980
1987	50,603	30,077
1988	52,174	31,920
1989	53,529	33,436
1990	53,887	35,079
1991	52,411	36,068
1992	53,288	36,184
1993	53,867	36,161
1994	55,361	36,362
1995	56,065	36,733
1996	57,259	37,962

Source: Penn World Table 6.1

To eliminate from these time series any underlying technological progress, this exercise will consider the two transformed series that appear for each year and country dividing GDP per worker by the GDP per worker in the USA and multiplying by 100. That is, we normalize for each year the GDP per worker in the USA to 100, and compute for Japan the GDP per worker as a percentage of that in the USA. The resulting series are called $(y_t^U)_{t=59}^{96}$ and $(y_t^J)_{t=59}^{96}$, where the first is just a list of 100s. We will take the view that the two transformed series are the relevant data on the two countries. The issue is how well the basic Solow model can explain these series and thereby how Japan was able to catch up the USA.

1. Compute the series $(y_t^U)_{t=59}^{96}$ and $(y_t^J)_{t=59}^{96}$ and plot each of them against years in a diagram. Give some qualitative description of the improvement in Japan's GDP.

In the following we take (or perhaps test) the view that each country can be described by the basic Solow model with country-specific parameter values, and that the USA has been in steady state in all the considered years. We assume that for both countries $\alpha = \frac{1}{3}$ and $\delta = 0.05$. For the savings and population growth rates of the USA we assume $s^U = 0.21$ and $n^U = 0.015$, respectively. For Japan we assume $s^J = 0.34$ and $n^J = 0.012$. These parameter values are realistic annual values (for some as averages for the considered period). We will first, in Questions 2 and 3, assume that the two countries had the same technological level B over the period considered.

2. What must B be for the USA to be in steady state at $y^{*U} = 100$? What are the steady state values, k^{*J} and y^{*J} , of capital and output per worker, respectively, in Japan with this B ? What must k_{59}^J (capital per worker in Japan in 1959) have been? Describe roughly how far Japan was from steady state in 1959 according to these computations (what were the percentages in 1959 of capital per worker and output per worker, respectively, to the steady state levels?).
3. Starting with the computed k_{59}^J , and assuming the computed B , simulate the model (the transition equation) for Japan over the period 1959 to 1996 and thereby create a model-based time series $(y_t^J)_{t=59}^{96}$ for GDP per worker in Japan. Plot the actual series, $(y_t^J)_{t=59}^{96}$, as well as the model-based one, $(y_t^J)_{t=59}^{96}$, against time in a diagram that also shows the steady state values, y^{*U} and y^{*J} , as horizontal lines. How fast has Japan actually caught up to the USA compared to how fast it should have caught up according to the Solow model under the assumption of a *common technological level*?

The assumption of the same B in the USA and Japan may be heroic. Another way to proceed in simulations is to rely on direct data for capital per worker in Japan in 1959 and compute an associated technological parameter B^J , which can be different from that in the USA (the B above). Data for capital are not the most reliable, but some empirical material suggests that in 1959 capital per worker in Japan was about one half of GDP per worker in Japan in the same year. In Questions 4 and 5 we therefore assume a level $k_{59}^{J2} = y_{59}^J/2$ of capital per worker in Japan in 1959.

4. What must B^J be? Compute the associated steady state values, k^{*J2} and y^{*J2} , for Japan. How far from steady state was Japan in 1959 according to these computations?
5. Simulate the model for Japan again, this time starting with k_{59}^{J2} and assuming the computed B^J , over the years 1959 to 1996, and thus create a new model-based series, $(y_t^J)_{t=59}^{96}$, for output per worker in Japan. Plot $(y_t^J)_{t=59}^{96}$ and $(y_t^J)_{t=59}^{96}$ against time in a diagram also indicating the steady state values, y^{*U} and y^{*J2} , as horizontal lines. Comment on the goodness of the model's fit and on the speed with which Japan actually caught up to the USA compared with the speed with which Japan should have caught up according to the Solow model under the assumption of *different technological levels*.

Exercise 11. Endogenous population growth and club convergence in the basic Solow model

Consider a growth model that consists of the same equations as the basic Solow model except that the n in Eq. (19) is no longer exogenous, but rather an endogenous variable that

depends on prosperity. Hence, in (19), n is replaced with n_t , and an additional equation, $n_t = h(y_t) = h(Bk_t^\alpha)$, or just $n_t = n(k_t)$, states how the population growth rate depends on income per capita and hence on the capital-labour ratio. Show that the Solow equation that can be derived from the model defined this way is:

$$k_{t+1} - k_t = \frac{1}{1 + n(k_t)} (sBk_t^\alpha - (n(k_t) + \delta)k_t),$$

and that the modified Solow equation is:

$$\frac{k_{t+1} - k_t}{k_t} = \frac{1}{1 + n(k_t)} [sBk_t^{\alpha-1} - (n(k_t) + \delta)].$$

It seems reasonable to assume a shape of the function $n(k_t)$ as illustrated in the figure below. For low values of k_t , the function $n(k_t)$ is perhaps negative but increasing, since higher k_t and y_t mean better living conditions, less infant mortality, etc. This is true for k_t up to a certain value. From that value, $n(k_t)$ is decreasing, because as people get richer birth rates fall for various reasons. However, as k_t increases from the limiting value, $n(k_t)$ never falls below a certain positive value, n , in the figure.

Show by an appropriate illustration in the modified Solow diagram that with such a shape of $n(k_t)$, there may be three steady state values for k_t , that is, three intersections between the curve $sBk_t^{\alpha-1}$, and the curve $n(k_t) + \delta$. If this is the case, how does the (unmodified) Solow diagram look? Which of the steady states are stable in the sense that if k_t departs a little from the steady state in question, the dynamics of the model tend to bring it back towards the steady state, and which one is unstable? Relate the curves you have drawn in the Solow diagram (unmodified or modified) to the idea of club convergence explained in Chapter 2. Explain intuitively why it is that if k_t (and hence y_t) starts below a certain threshold value it converges to a low steady state value, while if it starts above the threshold value it converges to a high steady state value.

